
Metric Spaces & Topology



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Caveat Lector: unofficial notes. Comments and corrections should be sent to kb514@ic.ac.uk. Other notes available at wwwf.imperial.ac.uk/~kb514.

Syllabus

This module extends various concepts from analysis to more general spaces.

Metric spaces: definition, examples including function spaces; continuity; boundedness; subspaces; open and closed subsets, continuity in terms of open sets; equivalence of metrics. [6]

Topological Spaces: definition, examples; finer and coarser topologies; continuity; bases; subspaces; homeomorphisms; neighbourhoods, closed sets, limit points, the closure of a set; dense sets. [6]

The Hausdorff Property. [1]

Compactness: compact subsets of metric spaces; compact subsets of Hausdorff spaces; compactness is a topological property; the Heine-Borel Theorem. [3]

Sequential Compactness: convergent sequences and Cauchy sequences; the Bolzano-Weierstrass Theorem; Lebesgue number. [2]

Complete metric spaces: completeness; total boundedness; convergence in function spaces: pointwise and uniform convergence; the Cauchy Criterion; the Contraction Mapping Theorem (aka Banachs Fixed Point Theorem) [3]

The Fundamental Group: paths and homotopies; definition of the fundamental group; the fundamental group of the circle; winding number application: the Fundamental Theorem of Algebra [6]

Appropriate books

W.A. Sutherland *Introduction to Metric and Topological Spaces*. Clarendon.
A. Hatcher *Algebraic Topology*. CUP.

Course website: coates.ma.ic.ac.uk/M2PM5

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1 Metric Spaces

Reminder: Given $f : \mathbb{R} \rightarrow \mathbb{R}$, what does it mean for f to be cts at a ?

Lecture 1

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Expand on this: Given $\epsilon > 0$, $\exists \delta > 0$ such that

$$|f(x) - f(a)| < \epsilon \text{ whenever } |x - a| < \delta$$

In words: f is continuous at $x = a \iff$ I can make $f(x)$ arbitrarily close to $f(a)$ by taking x sufficiently close to a .

What does it mean that for $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ to be continuous at $a \in \mathbb{R}^n$?

$$\lim_{\underline{x} \rightarrow \underline{a}} f(\underline{x}) = f(\underline{a})$$

where $\underline{x}, \underline{a}$ are vectors in $\mathbb{R}^n$, $f(\underline{a}), f(\underline{x})$ vectors in $\mathbb{R}^m$. Expand on this: Given $\epsilon > 0$, $\exists \delta > 0$ such that

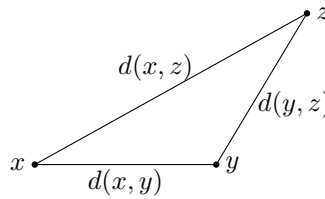
$$\|f(\underline{x}) - f(\underline{a})\| < \epsilon \text{ whenever } \|\underline{x} - \underline{a}\| < \delta$$

(where $\|\underline{x}\| = \sqrt{x_1^2 + \dots + x_n^2}$.)

Definition (Metric Space). A *metric space* is a pair (X, d) where X is a non-empty set and $d : X \times X \rightarrow \mathbb{R}$ is a function satisfying:

- (M1): $d(x, y) \geq 0$ for all $x, y \in X$, with equality iff $x = y$.
- (M2): $d(x, y) = d(y, x)$ for all $x, y \in X$. [Symmetry]
- (M3): $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y \in X$. [$\triangle$ inequality]

$\triangle$ inequality in $\mathbb{R}^2$:



Example 1.1 (Euclidean metric on $\mathbb{R}^n$). Take $X = \mathbb{R}^n$ and

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

where $\underline{x} = (x_1, \dots, x_n)^T$, $\underline{y} = (y_1, \dots, y_n)^T$.

Claim: d is a metric.

Proof. Check (M1) - (M3):

(M1) is obvious.

(M2) is *obvious*.

(M3): Put $z = (z_1, \dots, z_n)^T$. Set

$$r_i = x_i - y_i, s_i = y_i - z_i$$

We need to show that

$$d(x, z) = d(x, y) + d(y, z)$$

i.e.

$$\sqrt{\sum_{i=1}^n (x_i - z_i)^2} \leq \sqrt{\sum_{i=1}^n (x_i - y_i)^2} + \sqrt{\sum_{i=1}^n (y_i - z_i)^2}$$

i.e.

$$\sqrt{\sum_{i=1}^n (r_i + s_i)^2} \leq \sqrt{\sum_{i=1}^n r_i^2} + \sqrt{\sum_{i=1}^n s_i^2}$$

This is true if and only if

$$\sum_{i=1}^n (r_i + s_i)^2 \leq \left(\sqrt{\sum_{i=1}^n r_i^2} + \sqrt{\sum_{i=1}^n s_i^2} \right)^2$$

This is true if and only if

$$2 \sum_{i=1}^n r_i s_i \leq 2 \sqrt{\left(\sum_{i=1}^n r_i^2 \right) \left(\sum_{i=1}^n s_i^2 \right)}$$

This is the Cauchy-Schwartz inequality (**M1GLA**): it holds for all r_i, s_i .
So (M3) holds. Thus d is a metric. ■

Example 1.2 (Discrete metric). Take $X =$ any non-empty set and

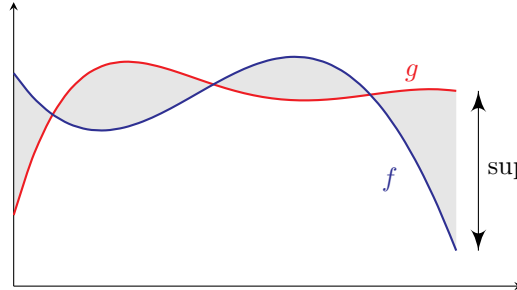
$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

(M1), (M2) obvious. (M3) left as an easy *exercise*.

Example 1.3. Take $X = \{\text{all continuous functions } f : [a, b] \rightarrow \mathbb{R}\}$, and

$$d(f, g) = \int_a^b |f(t) - g(t)| \, dt$$

E.g. $d = \text{shaded area}$:



Exercise: If $h : [a, b] \rightarrow \mathbb{R}$ is continuous and $h(t) \geq 0$ for all $t \in [a, b]$ then

$$\int_a^b h(t) \, dt \geq 0$$

with equality iff $h(t) = 0$ or all t .

Apply this with $h(t) = |f(t) - g(t)|$ which is continuous. Then $d(f, g) \geq 0$ with equality iff $|f(t) - g(t)| = 0$ for all $t \in [a, b]$, i.e. with equality iff $f = g$. So (M1) holds.

(M2): $d(f, g) = d(g, f)$ is obvious.

(M3): Let $f, g, h : [a, b] \rightarrow \mathbb{R}$ be continuous functions. Then

$$\begin{aligned} d(f, h) &= \int_a^b |f(t) - h(t)| \, dt \\ &= \int_a^b |f(t) - g(t) + g(t) - h(t)| \, dt \\ &\leq \int_a^b (|f(t) - g(t)| + |g(t) - h(t)|) \, dt \\ &= \int_a^b |f(t) - g(t)| \, dt + \int_a^b |g(t) - h(t)| \, dt \\ &= d(f, g) + d(g, h) \end{aligned}$$

So (M3) holds, hence (X, d) is a metric space.

Exercise: Take $X = \{\text{all continuous functions } f : [a, b] \rightarrow \mathbb{R}\}$ and show that

$$d(f, g) = \sup_{t \in [a, b]} |f(t) - g(t)|$$

is also a metric.

Lots more examples in textbook (on your reading.) Some examples can be

very strange looking, e.g. the discrete metric: $X = \text{any non-empty set}$,

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

A good source for counterexamples. Similar things can be found in *Counterexamples in Topology*, by Steen & Seebach [515.1 STE]

1.1 Continuity

Reminder: $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at $a \in \mathbb{R}$ iff given $\epsilon > 0$, $\exists \delta > 0$ such that if $|x - a| < \delta$ then $|f(x) - f(a)| < \epsilon$. We can generalise this:

Definition (Continuity). Let (X, d_X) and (Y, d_Y) be metric spaces and let $f : X \rightarrow Y$ be a function. We say f is *continuous* at $a \in X$ if and only if $\forall \epsilon, \exists \delta > 0$ such that whenever $d_X(x, a) < \delta$ we have $d_Y(f(x), f(a)) < \epsilon$.

We say that f is *continuous* iff f is continuous at $a \in X$ for all $a \in X$.

For $(X, d_X) = (\mathbb{R}^n, d)$ and $(Y, d_Y) = (\mathbb{R}^m, d)$ with $d = \text{Euclidean metric}$, then this is the same definition as in M1P1. So you already know many continuous functions.

Example 1.4. $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ as $(x, y) \mapsto x + y$ is continuous, $f : \mathbb{R} \rightarrow \mathbb{R}$ with $x \mapsto |x|$ is continuous, etc.

Proposition 1.5 (The composition of continuous functions is continuous.). Let (X, d_X) , (Y, d_Y) , (Z, d_Z) be metric spaces. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions such that f is continuous at $a \in X$ and g is continuous at $f(a) \in Y$. Then $g \circ f : X \rightarrow Z$ is continuous at a .

Proof. Fix $\epsilon > 0$. Since g is continuous at $f(a) \in Y$, we know $\exists \delta' > 0$ such that whenever $d_Y(y, f(a)) < \delta'$,

$$d_Z(g(y), g(f(a))) < \epsilon$$

Because f is continuous at $a \in X$, $\exists \delta > 0$ such that whenever $d_X(x, a) < \delta$,

$$d_Y(f(x), f(a)) < \delta'$$

Thus whenever $d_X(x, a) < \delta$, we have

$$d_Y(f(x), f(a)) < \delta'$$

now setting $y = f(x)$, we have

$$d_Z(gf(x), gf(a)) < \epsilon$$

Thus $g \circ f : X \rightarrow Z$ is continuous at $x = a$. ■

In particular, if $f : X \rightarrow Y$ is continuous and $g : Y \rightarrow Z$ is continuous, then $g \circ f : X \rightarrow Z$ is continuous.

Corollary 1.6. *Suppose that (X, d) is a metric space and $f, g : X \rightarrow \mathbb{R}$ are continuous functions¹. Then* Lecture 3

- (a) $|f| : X \rightarrow \mathbb{R}$ where $x \mapsto |f(x)|$
- (b) $f + g : X \rightarrow \mathbb{R}$ where $x \mapsto f(x) + g(x)$
- (c) $f \times g : X \rightarrow \mathbb{R}$, with $x \mapsto f(x)g(x)$

are all continuous.

Proof. The function $|\cdot| : \mathbb{R} \rightarrow \mathbb{R}$, where $y \mapsto |y|$ is continuous. So applying the previous proposition, the composition of continuous functions is continuous, so $|f|$ is continuous. (b) and (c) follow similarly, since we know that the function $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ sending $a, b \mapsto a + b$ is continuous, and the function $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ sending $a, b \mapsto ab$ is also continuous. ■

1.2 Product Spaces

Given two metric spaces (X, d_X) and (Y, d_Y) , can we put a metric on $X \times Y$? Yes; in lots of ways. We could define

$$d_1((x, y), (x', y')) = d_X(x, x') + d_Y(y, y')$$

or

$$d_2((x, y), (x', y')) = \sqrt{d_X(x, x')^2 + d_Y(y, y')^2}$$

or

$$d_p((x, y), (x', y')) = \sqrt[p]{d_X(x, x')^p + d_Y(y, y')^p}$$

for $1 \leq p < \infty$.

Showing that d_p is a metric comes down to **Hölder's inequality**:

$$\sum_{k=1}^n |a_k b_k| \leq \left(\sum_{k=1}^n |a_k|^p \right)^{1/p} \left(\sum_{k=1}^n |b_k|^q \right)^{1/q}$$

where $p, q \in [1, \infty)$ such that $1/p + 1/q = 1$.

What happens when $p \rightarrow \infty$?

$$d_\infty((x, y), (x', y')) = \max(d_X(x, x'), d_Y(y, y'))$$

Exercise: Show that these are all metrics on $X \times Y$.

Later we will see that all of these choices are equivalent in a precise sense. In particular, they all lead to the same notions of continuous functions and they lead to the same notion of convergent sequences. [They lead to the same topology on $X \times Y$]. Until then, always use d_1 on product spaces, unless otherwise stated.

¹Any time I write $\mathbb{R}^n$ we use the Euclidean metric.

Proposition 1.7. *Let $f : X \rightarrow X'$, $g : Y \rightarrow Y'$ be maps of metric spaces. Suppose that f is continuous at $a \in X$, g continuous at $b \in Y$. Then the map*

$$f \times g : X \times Y \rightarrow X' \times Y'$$

where $(x, y) \mapsto (f(x), g(y))$ is continuous at $(a, b) \in X \times Y$.

Proof. Let $d_X, d_Y, d_{X'}, d_{Y'}$ be the metrics on X, Y, X' and Y' respectively. The metric on $X \times Y$ is then

$$d_{X \times Y}((x_1, y_1), (x_2, y_2)) = d_X(x_1, x_2) + d_Y(y_1, y_2)$$

and similarly

$$d_{X' \times Y'}((x'_1, y'_1), (x'_2, y'_2)) = d_{X'}(x'_1, x'_2) + d_{Y'}(y'_1, y'_2)$$

We're trying to prove that $f \times g$ is continuous at $(a, b) \in X \times Y$, so fix $\epsilon > 0$. By continuity of f, g we know that $\exists \delta_1, \delta_2 > 0$ such that

$$d_{X'}(f(x), f(a)) < \epsilon/2 \quad (1)$$

whenever $d_X(x, a) < \delta_1$, and

$$d_{Y'}(g(y), g(b)) < \epsilon/2 \quad (2)$$

whenever $d_Y(y, b) < \delta_2$.

So set $\delta := \min(\delta_1, \delta_2)$. Then whenever $d_{X \times Y}((x, y), (a, b)) < \delta$, we have

$$d_X(x, a) + d_Y(y, b) < \delta$$

$\implies d_X(x, a) < \delta \leq \delta_1$ and $d_Y(y, b) < \delta \leq \delta_2$. Thus (1) and (2) holds, and so

$$d_{X' \times Y'}((f(x), g(y)), (f(a), g(b))) < \epsilon$$

So $f \times g$ is continuous at $(a, b) \in X \times Y$. ■

Proposition 1.8. *Let $(X, d_X), (Y, d_Y)$ be metric spaces. Then the projection maps*

$$p_X : X \times Y \rightarrow X, \quad (x, y) \mapsto x$$

and

$$p_Y : X \times Y \rightarrow Y, \quad (x, y) \mapsto y$$

are both continuous.

Proof. Omitted.

Definition. The diagonal map $\Delta : X \rightarrow X \times X$ sends $x \mapsto (x, x)$.

Proposition 1.9. *Let (X, d) be a metric space. Then the diagonal map $\Delta : X \rightarrow X \times X$ is continuous.*

To write down a continuous map $X \times Y \rightarrow X' \times Y'$, I can take $(x, y) \mapsto (f(x), g(y))$ where $f : X \rightarrow X'$, $g : Y \rightarrow Y'$ are continuous. Suppose I want to write down a continuous map $X \rightarrow X' \times Y'$.

Is $x \mapsto (f(x), g(x))$ continuous, given $f : X \rightarrow X'$, $g : Y \rightarrow Y'$ are? Yes!

We know

$$X \times X \longrightarrow X' \times Y'$$

with $(x, y) \mapsto (f(x), g(y))$ and

$$X \xrightarrow{\Delta} X \times X$$

with $x \mapsto (x, x)$ are both continuous. Then we can compose these:

$$X \longrightarrow X \times X \longrightarrow X' \times Y'$$

Subspaces

Recall the 3 properties that define a metric:

Lecture 4

- (M1): $d(x, y) \geq 0$ for all $x, y \in X$, with equality iff $x = y$.
- (M2): $d(x, y) = d(y, x)$ for all $x, y \in X$. [Symmetry]
- (M3): $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y \in X$. [Δ inequality]

If (X, d) is a metric space and $A \subseteq X$ is *any* non-empty subset of X , then (A, d) is also a metric. [technically $d|_A : A \times A \rightarrow \mathbb{R}$ sending $(a, a') \mapsto d(a, a')$].

Definition. (A, d) is called a *subspace* of (X, d) and the metric on A is called the *subspace metric*.

1.3 Boundedness

Definition (Bounded). A subset S of a metric space (X, d) is called *bounded* if and only if $S = \emptyset$ or there exists points $x_0 \in S$ and $K \in \mathbb{R}$ such that $d(x, x_0) \leq K$ for all $x \in S$.

Note that if this holds for some $x_0 \in S$, then it holds for *any* choice of $x_0 \in S$ (but for a different K):

Proof. Let $x_1 \in S$ be arbitrary. Then, for all $x \in S$, by the Δ inequality

$$d(x, x_1) \leq d(x, x_0) + d(x_0, x_1)$$

so, setting $K' = K + d(x_0, x_1)$, we see that

$$d(x, x_1) \leq K' \quad \forall x \in S$$

■

Note also that if S is a bounded subset of (X, d) then $d(x, y)$ is bounded for any $x, y \in S$, because S is bounded $\implies \exists x_0 \in S, K \in \mathbb{R}$ such that

$$d(x, x_0) \leq K \quad \forall x \in S$$

Let $x, y \in S$ be arbitrary. Then

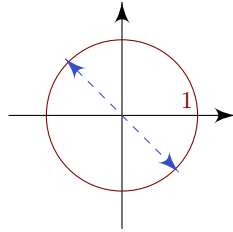
$$d(x, y) \leq d(x, x_0) + d(x_0, y) \leq 2K$$

So the following definition makes sense:

Definition. Let S be a non-empty bounded subset of a metric space (X, d) . The *diameter* of S is defined to be

$$\sup\{d(x, y) : x, y \in S\}$$

Example 1.10. Take $X = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ as a subspace of $\mathbb{R}^2$. Then the diameter of X is 2:



Convention. We define $\text{diameter}(\emptyset) = 0$.

Definition. Let S be a set, (X, d) be a metric space and $f : S \rightarrow X$ a map. We say that f is *bounded* iff the subset $f(S) \subseteq X$ is bounded.

So taking $X = \mathbb{R}$, $S = \mathbb{R}$ gives the usual notion of a bounded function $f : \mathbb{R} \rightarrow \mathbb{R}$.

Proposition 1.11. Let $S_1, S_2, \dots, S_n$ be bounded subsets of a metric space (X, d) . Then

$$\bigcup_{i=1}^n S_i$$

is also bounded.

Remark. This is not true for infinite sets, e.g. Take $X = \mathbb{R}$, $S_i = \{i\}$ for $i \in \mathbb{N}$. Then each S_i is bounded, but $\bigcup_i S_i$ is not.

Proof. It is sufficient to prove this for $n = 2$. (induction.)

Then $\exists K_1 \in \mathbb{R}$ such that $d(x, x_1) \leq K_1$ for all $x \in S_1$ and $\exists K_2 \in \mathbb{R}$ such that $d(x, x_2) \leq K_2$ for all $x \in S_2$. (Here $x_1 \in S_1$ and $x_2 \in S_2$ are chosen points.) We want to show that $S_1 \cup S_2$ is bounded. It suffices to show that $\exists K \in \mathbb{R}$ such that $d(x, x_1) \leq K$ for all $x \in S_1 \cup S_2$. Take

$$K = \max(K_1, K_2 + d(x_1, x_2))$$

Then either $x \in S_1$, in which case $d(x, x_1) \leq K_1 \leq K$, or $x \in S_2$, in which case $d(x, x_1) \leq d(x, x_2) + d(x_2, x_1) \leq K_2 + d(x_1, x_2) \leq K$. So $S_1 \cup S_2$ is bounded. ■

1.4 Open and Closed Sets

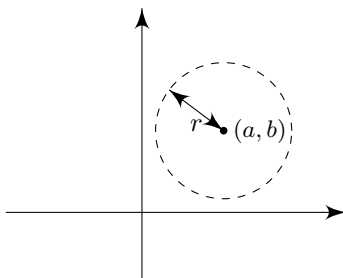
Let (X, d) be a metric space and $x_0 \in X$. Let $r > 0, r \in \mathbb{R}$.

Definition. The *open ball* in X of radius r centred at $x_0 \in X$ is

$$B_r(x_0) = \{x \in X : d(x_0, x) < r\}$$

Example 1.12. Take $X = \mathbb{R}^2$ and $x_0 = (a, b)$. Then

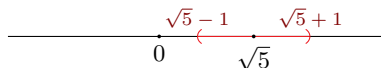
$$B_r(x_0) = \{(x, y) \in \mathbb{R}^2 : (x - a)^2 + (y - b)^2 < r^2\}$$



Example 1.13. Take $X = \mathbb{R}$ and $x_0 = \sqrt{5}$, $r = 1$. Then

$$B_1(\sqrt{5}) = \{x \in \mathbb{R} : |x - \sqrt{5}| < 1\}$$

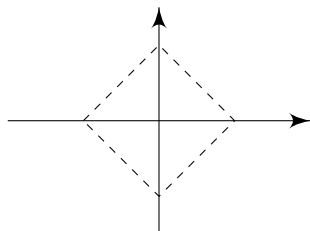
is



Example 1.14. Take $X = \mathbb{R}^2$ and the taxi-cab metric

$$d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + |y_1 - y_2|$$

then $B_r((0, 0))$ looks like

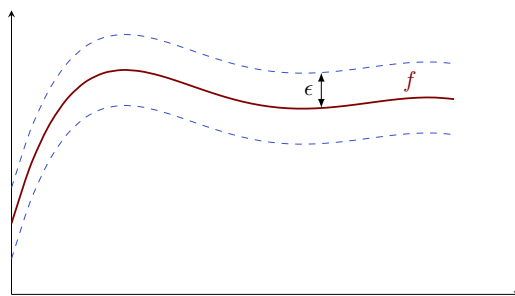


Balls don't have to be round!

Example 1.15. Take $X = \{f : [a, b] \rightarrow \mathbb{R} \text{ continuous}\}$ and

$$d(f, g) = \sup_{t \in [a, b]} |f(t) - g(t)|$$

What's $B_\epsilon(f)$? It's all the functions that lie within the dashed region:



i.e. $B_\epsilon(f) = \{g : [a, b] \rightarrow \mathbb{R} \text{ continuous s.t. } |f(t) - g(t)| < \epsilon \forall t \in [a, b]\}$

Lecture 5

Note: if $|f(t) - g(t)| < \epsilon$ for all $t \in [a, b]$, then

$$\sup_{t \in [a, b]} |f(t) - g(t)| \leq \epsilon$$

We need to show that this inequality is strict.

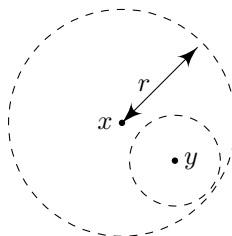
From M1P1: a continuous function $h : [a, b] \rightarrow \mathbb{R}$ is bounded and attains its bounds. Apply this with $h(t) = |f(t) - g(t)|$, and so

$$\sup_{t \in [a, b]} |f(t) - g(t)| < \epsilon$$

since this is equal to $|f(t_0) - g(t_0)|$ for some $t_0 \in [a, b]$.

Proposition 1.16. Given an open ball $B_r(x)$ in any metric space (X, d) and any $y \in B_r(x)$, there exists $\epsilon > 0$ such that $B_\epsilon(y) \subseteq B_r(x)$.

Proof. Idea in $\mathbb{R}^2$:



Fix $\epsilon = r - d(x, y) > 0$. I claim that $B_\epsilon(y) \subseteq B_r(x)$.

Let $z \in B_\epsilon(y)$ be arbitrary. We need to show $z \in B_r(x)$. But

$$\begin{aligned} d(x, z) &\leq d(x, y) + d(y, z) \\ &< d(x, y) + \epsilon \\ &= d(x, y) + r - d(x, y) = r \end{aligned}$$

Hence $z \in B_r(x)$. ■

Remark. The containment cannot be strict (since the discrete metric would not satisfy this).

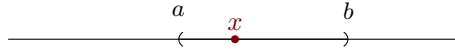
Definition (Open set). Let (X, d) be a metric space. A subset $U \subseteq X$ is called *open* iff for any $x \in U$, $\exists \epsilon > 0$ such that $B_\epsilon(x) \subseteq U$.

So the previous proposition says “open balls are open”.

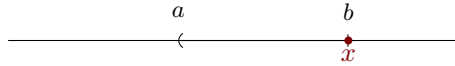
Example 1.17. Let (X, d) = any metric space. X is an open set. So is $\emptyset$.

Example 1.18. In $\mathbb{R}$, with Euclidean metric, we claim that for $a < b \in \mathbb{R}$:

- (a, b) is open.
- $(a, b]$ is not open.

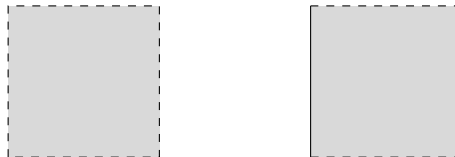


For any $x \in (a, b)$, take $\epsilon = \frac{1}{2} \min(b - x, b - a)$. Then $(x - \epsilon, x + \epsilon) \subseteq (a, b)$.



Any open ball centred at b contains a point $y : y > b$. So no open ball centered at b lies in $(a, b]$.

Example 1.19. In $\mathbb{R}^2$, $(a, b) \times (c, d)$ is open, but $[a, b] \times (c, d)$ is not:



Example 1.20. (X, d) is any non-empty set with the discrete metric. Any subset of X is open.

Proposition 1.21. Let $f : X \rightarrow Y$ be a map of metric spaces. Then f is continuous iff $f^{-1}(\mathcal{U})$ is open in X whenever $\mathcal{U}$ is open.

Proof. Suppose that f is continuous and that $\mathcal{U} \subseteq Y$ is open. We want to show that $f^{-1}(\mathcal{U})$ is open in X . Let $x \in f^{-1}(\mathcal{U})$ be arbitrary. We need to show that $\exists \delta > 0$ such that $B_\delta(x) \subseteq f^{-1}(\mathcal{U})$, or in other words, $f(B_\delta(x)) \subseteq \mathcal{U}$.

But $x \in f^{-1}(\mathcal{U})$, so $f(x) \in \mathcal{U}$. And $\mathcal{U}$ is open, so $\exists \epsilon > 0$ such that $B_\epsilon(f(x)) \subseteq \mathcal{U}$. Now f is continuous at $x \in X$, so $\exists \delta > 0$ such that $d_Y(f(x), f(x')) < \epsilon$ whenever $d_X(x, x') < \delta$. In other words, $f(B_\delta(x)) \subseteq B_\epsilon(f(x)) \implies f(B_\delta(x)) \subseteq \mathcal{U}$. So $B_\delta(x) \subseteq f^{-1}(\mathcal{U})$, as required. $\square$

Conversely, suppose that $f^{-1}(\mathcal{U})$ is open in X whenever $\mathcal{U}$ is open in Y . We need to show that f is continuous. Let $x_0 \in X$ be arbitrary. It suffices to show that f is continuous at x_0 . In other words, it suffices to show that for all $\epsilon > 0$, $\exists \delta > 0$ such that

$$d_Y(f(x), f(x_0)) < \epsilon \text{ whenever } d_X(x, x_0) < \delta$$

Fix $\epsilon > 0$, and consider $\mathcal{U} = B_\epsilon(f(x_0))$. $\mathcal{U}$ is open, so by assumption, $f^{-1}(\mathcal{U})$ is open. We know that $x_0 \in f^{-1}(\mathcal{U})$, so $\exists \delta > 0$ such that $B_\delta(x_0) \subseteq f^{-1}(\mathcal{U})$. So whenever $d(x, x_0) < \delta$ we have $f(x) \in \mathcal{U}$, i.e. whenever $d(x, x_0) < \delta$ we have $d(f(x), f(x_0)) < \epsilon$. Thus f is continuous at $x_0 \in X$ and we're done. $\blacksquare$

Remark. $f^{-1}(\mathcal{U})$ being open in X whenever $\mathcal{U}$ is open in Y does NOT mean that f takes open sets to open sets. E.g. $f : \mathbb{R} \rightarrow \mathbb{R}$ where $x \mapsto 0$ is certainly continuous, so $f^{-1}(\mathcal{U})$ is open in $\mathbb{R}$ whenever $\mathcal{U}$ is open in $\mathbb{R}$, but every non-empty set maps to $\{0\}$.

Exercise: Think of another continuous map $f : X \rightarrow Y$ of metric spaces and $\mathcal{U} \subset X$ an open set such that $f(\mathcal{U})$ is not open.

Proposition 1.22.

- (i) Let $\mathcal{U}_1, \dots, \mathcal{U}_n$ be open sets in a metric space (X, d) . Then the intersection $\bigcap_{i=1}^n \mathcal{U}_i$ is also open.
- (ii) Let $\mathcal{U}_i, i \in I$ be open sets in a metric space (X, d) . Then $\bigcup_{i \in I} \mathcal{U}_i$ is open in (X, d) .

Proof. (i) Let $x \in \bigcap_{i=1}^n \mathcal{U}_i$ be arbitrary. We need to show that $\exists \epsilon > 0$ such that $B_\epsilon(x) \subseteq \bigcap_{i=1}^n \mathcal{U}_i$. Now $x \in \mathcal{U}_i$ for each i and $\mathcal{U}_i$ is open, so $\exists \epsilon_i > 0$ such that $B_{\epsilon_i}(x) \subseteq \mathcal{U}_i$. So set $\epsilon = \min(\epsilon_1, \dots, \epsilon_n) > 0$. Then $B_\epsilon(x) \subseteq \mathcal{U}_i$ for each i , and so

$$B_\epsilon(x) \subseteq \bigcap_{i=1}^n \mathcal{U}_i$$

(ii) is left as an *exercise*. $\blacksquare$

Definition (Closed set). A subset V of a metric space (X, d) is *closed* if and only if $X \setminus V$ is open.

Examples 1.23.

$$V = \text{[shaded square]} \subset \mathbb{R}^2$$

i.e. $V : |x| \leq a, |y| \leq b$ is closed, because $\mathbb{R}^2 \setminus V$ is open.

$$V = \text{[shaded square with dashed border]} \subset \mathbb{R}^2$$

i.e. $V : |x| \leq a, |y| < b$ is not closed, because $\mathbb{R}^2 \setminus V$ is not open.



Warning. As in the previous example, closed is *not* the opposite of open. Sets are not doors; they can be both or neither open and closed.

Example 1.24. $\mathbb{N} \subset \mathbb{R}$ is closed, since $\mathbb{R} \setminus \mathbb{N}$ is open.

Example 1.25. Let $X =$ any non-empty set and $d =$ discrete metric. Any $\mathcal{U} \subset X$ is open. Any $V \subseteq X$ is closed.

Proposition 1.26.

- (i) Let $V_1, \dots, V_n$ be closed subsets of a metric space (X, d) . Then $\bigcup_{i=1}^n V_i$ is closed.
- (ii) Let $V_i, i \in I$ be closed subsets of X . Then $\bigcap_{i \in I} V_i$ is closed.

Proof. By De Morgan's laws:

$$X \setminus \left(\bigcup_{i \in I} X_i \right) = \bigcap_{i \in I} (X \setminus X_i)$$

and

$$X \setminus \left(\bigcap_{i \in I} X_i \right) = \bigcup_{i \in I} (X \setminus X_i)$$

So this follows from the previous proposition about open sets. ■

Notice that closed sets contain their limit points, i.e.

$$\begin{aligned} V \text{ closed} &\iff X \setminus V \text{ open} \\ &\iff \text{every point of } X \setminus V \text{ is surrounded by a small ball of points in } X \setminus V \\ &\iff \text{no points of } X \setminus V \text{ has a sequence of points of } V \text{ leading to it} \end{aligned}$$

Definition (Limit Point). Let (X, d) be a metric space and $A \subseteq X$. A point $x \in X$ is called a *limit point* of A (also called a *point of accumulation* of A) iff for all $\epsilon > 0$, $\exists y \in A$ with $y \in B_\epsilon(x)$ and $y \neq x$.

Example 1.27. Take $X = \mathbb{R}$ and $A = (0, 1] \cup \{2\}$. Then the limit points are $[0, 1]$.

Definition (Closure). Let (X, d) be a metric space and $A \subseteq X$ a subset of X . The *closure* $\bar{A}$ of A is

$$\bar{A} = \bigcap_{\substack{Z \subseteq X \\ Z \text{ closed}, A \subseteq Z}} Z$$

So $\bar{A}$ is the smallest closed subset of X that contains A . If Y is a closed subset of X and $A \subseteq Y$, then $\bar{A} \subseteq Y$.

Proposition 1.28. $\bar{A} = A \cup \{\text{limit points of } A\}$.

Proof. Set $B = A \cup \{\text{limit points of } A\}$. We need to show

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- (i) B is closed.
- (ii) Any closed subset V of X such that $A \subseteq V$ also contains all limit points of A .

(i) Let $\mathcal{U} = X \setminus B$. We need to show that $\mathcal{U}$ is open. Let $x \in \mathcal{U}$ be arbitrary. We need to show that $\exists \epsilon > 0$ such that $B_\epsilon(x) \subseteq \mathcal{U}$. Suppose not. ($\epsilon = \frac{1}{2n}$). Then $\exists a_n \in B$ such that $d(x, a_n) < \frac{1}{2n}$. Either $a_n \in A$, in which case set $x_n = a_n$, or a_n is a limit point of A , in which case we can find a point x_n of A with $d(x_n, a_n) < \frac{1}{2n} \implies d(x_n, x) < \frac{1}{2n} + \frac{1}{2n} = \frac{1}{n}$.

So I've found a sequence $\{x_n\}_{n \geq 1}$ of points of A such that $d(x_n, x) < 1/n$, so x is a limit point of A , ✖ since $x \in X \setminus B$ and B contains all the limit points of A . So $\exists \epsilon > 0$ such that $B_\epsilon(x) \subseteq \mathcal{U} \implies \mathcal{U}$ is open $\implies B$ is closed.

(ii) Let $Z \subseteq X$ be closed and suppose that $A \subseteq Z$. We need to show that Z contains all the limit points of A . Let $x \in X$ be a limit point of A . We need to show $x \in Z$. Suppose not. Then $x \in X \setminus Z$, which is open. So $\exists \epsilon > 0$ such that $B_\epsilon(x) \subseteq X \setminus Z$. But x is a limit point of A , so $\exists y \in B_\epsilon(x)$ such that $y \in A$ and $y \neq x$, ✖ since $B_\epsilon(x) \subseteq X \setminus Z$ and $A \subseteq Z$, so y can't be in $B_\epsilon(x)$. Therefore $x \in Z$, as required. ■

Example 1.29. Let

$$A = \boxed{} \subset \mathbb{R}^2$$

$$|x| \leq a, |y| < b.$$

I claim that

$$\overline{A} = \boxed{} \subset \mathbb{R}^2$$

$$|x| \leq a, |y| \leq b.$$

Example 1.30. $\mathbb{Q} \subset \mathbb{R}$. What is $\overline{\mathbb{Q}}$? $\overline{\mathbb{Q}} = \mathbb{R}$. $(\overline{\mathbb{R} \setminus \mathbb{Q}}) = \mathbb{R}$.

Proposition 1.31. Let A, B be subsets of a metric space X . Then

- (i) $A \subseteq \overline{A}$.
- (ii) $A \subseteq B \implies \overline{A} \subseteq \overline{B}$.
- (iii) A is closed $\iff A = \overline{A}$.
- (iv) $\overline{\overline{A}} = \overline{A}$.
- (v) $\overline{A}$ is closed in X .
- (vi) $\overline{A}$ is the smallest closed subset of X that contains A .

Proof. Left as a straightforward exercise. ■

Interior

The opposite notion; the interior is the biggest open set contained in A .

Definition (Interior). Let A be a subset of a metric space (X, d) . The interior of A is $\mathring{A} = \bigcup_{\mathcal{U} \text{ open}} \mathcal{U}$. Equivalently

$$\mathring{A} = \{a \in A : \exists \epsilon > 0 \text{ such that } B_\epsilon(a) \subseteq A\}$$

Examples 1.32.

- $A = [a, b] \subset \mathbb{R}$. Then $\mathring{A} = (a, b)$.
- Interior of $(a, b]$ is also (a, b) .
- $\mathbb{Q} \subset \mathbb{R}$, then $\mathring{\mathbb{Q}} = \emptyset$.

Proposition 1.33. Let A and B be subsets of a metric space (X, d) . Then:

- (i) $\mathring{A} \subseteq A$.
- (ii) $A \subseteq B \implies \mathring{A} \subseteq \mathring{B}$.
- (iii) A is open in $X \iff A = \mathring{A}$.
- (iv) $\mathring{\mathring{A}} = \mathring{A}$.

(v) $\bar{A}$ is open in X .

(vi) $\bar{A}$ is the largest open subset of A [i.e. if $\mathcal{U} \subseteq A$ and $\mathcal{U}$ is open, then $\mathcal{U} \subseteq \mathring{A}$.]

Proof. $\mathring{A} = X \setminus (\overline{X \setminus A})$. Now look at the previous proposition. ■

Boundary

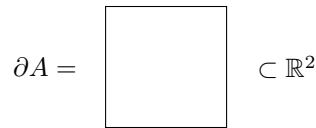
Definition (Boundary). The *boundary* of a subset A of a metric space (X, d) is $\partial A = \bar{A} \setminus \mathring{A}$.

Example 1.34. Take $X = \mathbb{R}$, $A = [a, b]$. Then $\bar{A} = [a, b]$ and $\mathring{A} = (a, b)$ so $\partial A = \{a, b\}$.

Example 1.35.



Then



$$|x| \leq a, |y| \leq b.$$

1.5 Convergence in Metric Spaces

The same notions we have in M1P1:

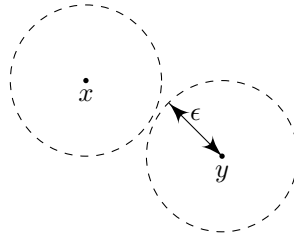
Lecture 8

Definition (Convergence). Let $(x_n)_{n \geq 1}$ be a sequence of points in a metric space (X, d) . We say that $x_n \rightarrow x$, where $x \in X$, given $\epsilon > 0$, $\exists N$ such that whenever $n \geq N$, we have $d(x, x_n) < \epsilon$.

Definition (Cauchy). A sequence $(x_n)_{n \geq 1}$ of points in a metric space (X, d) is called Cauchy sequence iff $\forall \epsilon > 0$, $\exists N$ such that whenever $n, m \geq N$, we have $d(x_n, x_m) < \epsilon$.

Proposition 1.36 (Uniqueness of limits). Let $(x_n)_{n \geq 1}$ be a sequence of points in a metric space (X, d) . Suppose that $x, y \in X$ are such that $x_n \rightarrow x$ and $x_n \rightarrow y$. Then $x = y$.

Picture:



Proof. Suppose $x \neq y$. Fix $\epsilon = \frac{1}{2}d(x, y) > 0$. Since $x_n \rightarrow x$, $\exists N_1$ such that $d(x_n, x) < \epsilon$ whenever $n \geq N_1$. Similarly since $x_n \rightarrow y$, $\exists N_2$ such that $d(x_n, y) < \epsilon$ whenever $n \geq N_2$.

Take $N = \max(N_1, N_2)$. Then

$$d(x, y) \leq d(x, x_N) + d(y, x_N) < 2\epsilon = d(x, y) \quad \text{✗}$$

So our assumption that $x \neq y$ must be false, thus $x = y$. ■

Proposition 1.37. Suppose that A is a subset of a metric space (X, d) and (a_n) is a sequence of points of A such that $a_n \rightarrow a \in X$. Then $a \in \bar{A}$.

Proof. If $a_n = a$ for some n , then there's nothing to prove (since $a_n \in A \subseteq \bar{A}$). Otherwise, $a_n \neq a$ for all n and for all $\epsilon > 0$, $\exists N$ such that whenever $n \geq N$ we have $a_n \in B_\epsilon(a)$. So a is a limit point of A . Thus $a \in \bar{A}$. ■

Exercise: Show any point in $\bar{A}$ is the limit of a sequence (a_n) of points in A .

Equivalent metrics - see reading pp. 61-72.



2 Topological Spaces

Motivation:

“shape” of a metric space $(X, d) \longrightarrow$ continuous functions on (X, d)

These only feel the open sets in (X, d) , not d itself.

Definition. A *topological space* $(X, \mathcal{T})$ consists of a non-empty set X together with a collection $\mathcal{T}$ of subsets of X such that

- (T1) $\emptyset \in \mathcal{T}, X \in \mathcal{T}$.
- (T2) If $\mathcal{U}, V \in \mathcal{T}$, then $\mathcal{U} \cap V \in \mathcal{T}$.
- (T3) If $\mathcal{U}_i \in \mathcal{T}, i \in I$ then $\bigcup_{i \in I} \mathcal{U}_i \in \mathcal{T}$.

Elements in $\mathcal{T}$ are called *open sets* in $(X, \mathcal{T})$. $\mathcal{T}$ is called a *topology* on X .

Example 2.1 (Metric Topology). Let (X, d) be any metric space. Let

$$\mathcal{T} = \{\mathcal{U} \subseteq X : \mathcal{U} \text{ is open w.r.t. } d\}$$

Then $\mathcal{T}$ satisfies (T1), (T2) & (T3), i.e. $\mathcal{T}$ is a topology. This is called the *metric topology* on X .

Example 2.2. Take $X = [0, 1], \mathcal{T} = \{\emptyset, X, [0, \frac{1}{2}]\}$ is a topology.

In general if we take $X =$ any non-empty sets, $\mathcal{T} = \{\emptyset, X, V\}$ where V is some fixed subset of X is a topology.

Example 2.3 (Euclidean Topology). Take $\mathbb{R}^n$ and

$$\mathcal{T} = \{\text{subsets of } \mathbb{R}^2 \text{ which are open w.r.t Euclidean metric}\}$$

this is just the metric topology on $\mathbb{R}^n$ induced by the Euclidean metric.

Example 2.4 (Discrete Topology). Take $X =$ any non-empty set, $\mathcal{T} = \{\text{all subsets of } X\}$ is a topology, called the *discrete topology*. This is the metric topology on X induced by the discrete metric.

Example 2.5 (Indiscrete Topology). Take $X =$ any non-empty set and $\mathcal{T} = \{\emptyset, X\}$ is a topology, but not a metric topology if X has more than one-point.

Exercise: Take $X =$ any non-empty set and $\mathcal{T} = \{\emptyset\} \cup \{\mathcal{U} \subseteq X : X \setminus \mathcal{U} \text{ is finite}\}$. Show that this is a topology on X . (The cofinite topology).

Proposition 2.6. Let X be a topological space. Then $\mathcal{U} \subseteq X$ is open iff for all $x \in \mathcal{U}$, there exists an open set N_x with $x \in N_x$ and $N_x \subseteq \mathcal{U}$. Lecture 9

Proof. Suppose that $\mathcal{U}$ is open. Then take $N_x = \mathcal{U}$ for all $x \in \mathcal{U}$. This works. Suppose, on the other hand, that for all $x \in \mathcal{U}$, $\exists$ open set N_x such that $x \in N_x$ and $N_x \subseteq \mathcal{U}$. Then

$$\mathcal{U} = \bigcup_{x \in \mathcal{U}} N_x$$

So by (T3), we have that $\mathcal{U}$ is open. ■

Definition (Coarser, Finer). Given two topologies $\mathcal{T}_1, \mathcal{T}_2$ on the same set X , we say that $\mathcal{T}_1$ is *coarser* than $\mathcal{T}_2 \iff \mathcal{T}_1 \subseteq \mathcal{T}_2$ and $\mathcal{T}_1$ is *finer* than $\mathcal{T}_2 \iff \mathcal{T}_2 \subseteq \mathcal{T}_1$.

Definition (Continuity). Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces and let $f : X \rightarrow Y$ be a map. We say that f is *continuous* if and only if $f^{-1}(\mathcal{U}) \in \mathcal{T}_X$ whenever $\mathcal{U} \in \mathcal{T}_Y$.

So if X and Y are metric spaces, and $\mathcal{T}_X$ and $\mathcal{T}_Y$ are the metric topologies, this is the *same* notion of continuous functions as before.

Aside: What about continuity at $x \in X$?

Definition. f is continuous at $x \in X$ iff for all $\mathcal{U} \in \mathcal{T}_Y$ such that $f(x) \in \mathcal{U}$, we have that $f^{-1}(\mathcal{U}) \in \mathcal{T}_X$.

But this turns out to not be so important for us. We are more concerned with global properties of shape.

Proposition 2.7. Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y), (Z, \mathcal{T}_Z)$ be topological spaces. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be continuous functions. Then $g \circ f : X \rightarrow Z$ is continuous.

Proof. Let $\mathcal{U} \in \mathcal{T}_Z$. We need to show that $(g \circ f)^{-1}(\mathcal{U}) \in \mathcal{T}_X$. But

$$(g \circ f)^{-1}(\mathcal{U}) = f^{-1}(g^{-1}(\mathcal{U}))$$

and $g^{-1}(\mathcal{U}) \in \mathcal{T}_Y$ by the continuity of g , so $f^{-1}(g^{-1}(\mathcal{U})) \in \mathcal{T}_X$ by the continuity of f . ■

Examples 2.8.

- (i) Let X be any topological space. Then $\text{id}: X \rightarrow X$ is continuous.
- (ii) Any constant map is continuous. In other words, Let X, Y be any topological spaces and let $f : X \rightarrow Y$ be the map sending $x \mapsto y_0$ for some fixed $y_0 \in Y$. Then f is continuous.

Proof. Let $\mathcal{U} \subseteq Y$ be open. Then

$$f^{-1}(\mathcal{U}) = \begin{cases} X & y_0 \in \mathcal{U} \\ \emptyset & y_0 \notin \mathcal{U} \end{cases}$$

So $f^{-1}(\mathcal{U})$ is open. ■

- (iii) Let X be equipped with the discrete topology, and let Y be any topological space, and let $f : X \rightarrow Y$ be any map. Then f is continuous.

Proof. Let $\mathcal{U} \subseteq Y$ be open. We need to show that $f^{-1}(\mathcal{U})$ is open in X . But all subsets of X are open. So this is true. ■

- (iv) Let X be any topological space. Let Y be equipped with the indiscrete topology. Let $f : X \rightarrow Y$ be any map. Then f is continuous.

Proof. The open sets in Y are $\emptyset, Y$. Now $f^{-1}(\emptyset) = \emptyset \in \mathcal{T}_X$ and $f^{-1}(Y) = X \in \mathcal{T}_X$. Therefore f is continuous. ■

Homeomorphisms

What does it mean for two topological spaces to be “the same” or “equivalent”?

Definition. A *homeomorphism* between topological spaces X and Y is a bijection $f : X \rightarrow Y$ such that f and f^{-1} are continuous.

Exercise: A bijection of sets $f : X \rightarrow Y$ gives a homeomorphism of topological spaces $X \rightarrow Y$ if and only if it induces a bijection $\mathcal{T}_X \rightarrow \mathcal{T}_Y$ given by sending $\mathcal{U} \mapsto f(\mathcal{U})$.

2.1 Bases

In metric spaces, open sets were unions of open balls. *What’s the analogue of this statement for topological spaces?*

Definition. Let $(X, \mathcal{T})$ be a topological space. A *basis* for $\mathcal{T}$ is a subfamily $\mathcal{B} \subseteq \mathcal{T}$ such that every set in $\mathcal{T}$ is a union of open sets in $\mathcal{B}$. We say $(X, \mathcal{T})$ has a *countable basis* for the topology if and only if $\exists$ a *countable basis*.

Example 2.9. Let $X = \mathbb{R}^2$ with the usual Euclidean topology. Then

$$\mathcal{B} = \{B_\epsilon(x) : \epsilon > 0, x \in \mathbb{R}^2\}$$

is a basis. Also

$$\mathcal{B}' = \{B_\epsilon(x) : \epsilon > 0, \epsilon \in \mathbb{Q}, x \in \mathbb{Q}^2\}$$

is a basis.

Aside: How can we get f such that f is continuous, f^{-1} not?

Lecture 10

Take X to discrete topology, Y indiscrete topology; then the identity map satisfies this: Suppose X has at least 2 points in it. Clearly $id : X \rightarrow Y$ is continuous, but $id : Y \rightarrow X$ is not; take any point not Y or $\emptyset$.

2.2 Closed Sets

Definition. A subset V of a topological space X is *closed* iff $X \setminus V$ is open.

Proposition 2.10. Let X be a topological space. Then:

- (C1) $X, \emptyset$ are closed in X .
- (C2) If V_1, V_2 are closed subsets of X , then $V_1 \cup V_2$ is closed.
- (C3) If $V_i, i \in I$ are such that V_i are closed, then $\bigcap_{i \in I} V_i$ is also closed.

Proof. Take compliments: Set $\mathcal{U}_i = X \setminus V_i$, then (C1), (C2), (C3) become (T1), (T2), (T3). ■

Example 2.11 (Euclidean Topology on $\mathbb{R}^2$).

$$A = \text{[dashed square]} \subset \mathbb{R}^2$$

$|x| \leq a, |y| < b$ is not closed. But

$$A = \text{[solid square]} \subset \mathbb{R}^2$$

$|x| \leq a, |y| \leq b$ is closed.

2.3 Closure

The closure of a subset A of a topological space X is the smallest closed subset of X that contains A . Precisely:

Definition. Let X be a topological space and $A \subseteq X$. The *closure* of A is

$$\overline{A} = \bigcap_{\substack{V \subseteq X \\ V \text{ closed, } A \subseteq V}} V$$

Example 2.12. See previous examples in metric spaces.

Example 2.13. Take X = any non-empty set with the indiscrete topology. If $A \subseteq X$ is any non-empty set, then $\overline{A} = X$.

Definition (Point of closure). Let X be a topological space and $A \subseteq X$. Then a point $x \in X$ is a *point of closure* of A iff every open set $\mathcal{U}$ such that $x \in \mathcal{U}$ satisfies $\mathcal{U} \cap A \neq \emptyset$.

Proposition 2.14. $\bar{A} = \{x \in X : x \text{ is a point of closure of } A\}$.

Proof. Set $B = \{x \in X : x \text{ is a point of closure of } A\}$. We want to show that $\bar{A} = B$. It suffices to show

(i) B is closed.

(ii) If V is a closed subset of X such that $A \subseteq V$, then $B \subseteq V$.

(i). Set $\mathcal{U} = X \setminus B$. We need to show $\mathcal{U}$ is open. Take $x \in \mathcal{U}$. We need to show that $\exists$ open set N_x such that $x \in N_x$ and $N_x \subseteq \mathcal{U}$. Suppose not. Then every open set N_x with $x \in N_x$ satisfies $N_x \not\subseteq \mathcal{U}$, i.e. $N_x \cap B \neq \emptyset$. So $\exists b \in N_x$ such that b is a point of closure of A . But N_x is an open set containing b , so N_x contains a point of A . Thus every open set N_x such that $x \in N_x$ contains a point of A . Thus x is a point of closure of A , i.e. $x \in B$, $\mathbb{X}$. So the assumption is false, i.e. $\exists N_x$ open such that $x \in N_x$ and $N_x \subseteq \mathcal{U}$. So $\mathcal{U}$ is open.

(ii). Suppose that V is closed in X and $A \subseteq V$. Let $b \in B$ be arbitrary. We need to show that $b \in V$. Suppose not. Then $b \in X \setminus V$ which is open. So $\exists \mathcal{U}$ open such that $b \in \mathcal{U}$ and $\mathcal{U} \subseteq X \setminus V$. Then $\mathcal{U} \cap A = \emptyset$, $\mathbb{X}$ since b is a point of closure with A , so any open set $\mathcal{U}$ with $b \in \mathcal{U}$ meets A . Thus $b \in V$ as required. ■

Proposition 2.15. Let A, B be subsets of a topological space X . Then:

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(i) $A \subseteq \bar{A}$.

(ii) $A \subseteq B \implies \bar{A} \subseteq \bar{B}$.

(iii) A is closed in $X \iff \bar{A} = A$.

(iv) $\bar{\bar{A}} = \bar{A}$.

(v) $\bar{A}$ is closed in X .

(vi) $\bar{A}$ is the smallest closed subset containing A .

Proof. Left as an exercise.

Interior

Definition. Let X be a topological space and A be a subset of X . Then $a \in X$ is called an *interior point* of A iff $\exists$ an open set N_a such that $a \in N_a$ and $N_a \subseteq A$. The *interior* of A is

$$\overset{\circ}{A} = \{a \in X : a \text{ is an interior point of } A\}$$

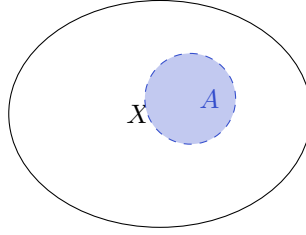
Equivalently

$$\overset{\circ}{A} = \bigcup_{\substack{\mathcal{U} \subseteq A \\ \mathcal{U} \text{ open}}} \mathcal{U}$$

Examples 2.16.

- $\mathbb{R}^2$ with the Euclidean topology, if $B = \{x \in \mathbb{R}^2 : |x| \leq 1\}$, then $\mathring{B} = \{x \in \mathbb{R}^2 : |x| < 1\}$.
- X = any non-empty set with the indiscrete topology, $A \subseteq X$ any non-empty subset of X . Then $\mathring{A} = \emptyset$ and $\overline{A} = X$. (cf. $\mathbb{Q} \subset \mathbb{R}$ with the Euclidean topology)

Proposition 2.17. $\overline{X \setminus A} = X \setminus \mathring{A}$.



Proof.

$$\begin{aligned}
 x \in \overline{X \setminus A} &\iff x \text{ is a point of closure of } X \setminus A \\
 &\iff \text{for all open sets } \mathcal{U} \text{ such that } x \in \mathcal{U}, \text{ we have } \mathcal{U} \cap (X \setminus A) \neq \emptyset \\
 &\iff \text{there is no open set } \mathcal{U} \text{ such that } x \in \mathcal{U} \text{ and } \mathcal{U} \subseteq A \\
 &\iff x \in \mathring{A}
 \end{aligned}$$

So $\overline{X \setminus A} = X \setminus \mathring{A}$. ■

Corollary 2.18. Let A, B be subsets of a topological space X . Then

- (i) $\mathring{A} \subseteq A$.
- (ii) If $A \subseteq B$ then $\mathring{A} \subseteq \mathring{B}$.
- (iii) A is open in $X \iff \mathring{A} = A$.
- (iv) $\mathring{\mathring{A}} = \mathring{A}$.
- (v) $\mathring{A}$ is open in X .
- (vi) $\mathring{A}$ is the largest open set contained in A .

Proof. Apply the previous proposition. This turns (i) - (vi) into the corresponding statements about closures. ■

Definition. The *boundary* of A is $\partial A = \overline{A} \setminus \mathring{A}$.

Useful Terminology. Let X be a topological space and $x \in X$. An *open neighbourhood* of x is an open set $\mathcal{U} \subseteq X$ such that $x \in \mathcal{U}$.

2.4 Subspaces

Definition. Let $(X, \mathcal{T})$ be a topological space and let A be a subset of X . The *subspace topology* on A is $\mathcal{T}_A = \{A \cap \mathcal{U} : \mathcal{U} \in \mathcal{T}\}$. A equipped with this topology is called a *subspace* of X .

Claim. $\mathcal{T}_A$ is a topology on A .

Proof. We need to show (T1) – (T3):

- $\emptyset \in \mathcal{T}_A$ because $\emptyset \in \mathcal{T}_A \implies A \cap \emptyset \in \mathcal{T}_A$.
- $A \in \mathcal{T}_A$ because $X \in \mathcal{T}_X \implies A \cap X \in \mathcal{T}_A$.
- Suppose that $\mathcal{U}, V \in \mathcal{T}_A$. Then $\mathcal{U} = \mathcal{U}' \cap A$ for some $\mathcal{U}' \in \mathcal{T}_A$ and $V = V' \cap A$ for some $V' \in \mathcal{T}$, so $\mathcal{U} \cap V = \mathcal{U}' \cap A \cap V' \cap A = (\mathcal{U}' \cap V') \cap A$.

■

Example 2.19. $A = [0, 1]$ and $X = \mathbb{R}$ with Euclidean topology. Then $(1 - c, 1]$ is open in the subspace topology.

Definition. The function $i : A \rightarrow X$ which sends $a \mapsto a$ is called the inclusion map, denoted i (or i_A)

Lecture 12

Lemma 2.20. Given $(X, \mathcal{T})$ a topological space, $A \subset X$ with the subspace topology, $i : A \rightarrow X$ is continuous.

Proof. We want for any $\mathcal{U} \in \mathcal{T}$, for $i^{-1}(\mathcal{U}) \in \mathcal{T}_A$. But for any $\mathcal{U} \in \mathcal{T}$, $i^{-1}(\mathcal{U}) = \mathcal{U} \cap A \in \mathcal{T}_A$ by definition. ■

Proposition 2.21. If $\mathcal{T}'$ is a topological space on A and $i : A \rightarrow X$ is continuous, then $\mathcal{T}_A \subset \mathcal{T}'$

Proof. If i continuous $\implies \forall \mathcal{U} \in \mathcal{T}, i^{-1}(\mathcal{U}) \in \mathcal{T}'$. But if $V \in \mathcal{T}_A$, $V = \mathcal{U} \cap A$ for some $\mathcal{U} \in \mathcal{T}$, so $V \in \mathcal{T}' \implies \mathcal{T}_A \subset \mathcal{T}'$. ■

Proposition 2.22. Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ be topological spaces as a $f : X \rightarrow Y$. Then f continuous $\implies f \circ i$ continuous.

$$\begin{array}{ccc} Y & \xleftarrow{f} & X \\ & \nwarrow f \circ i & \uparrow i \\ & & A \end{array}$$

Proof. The composition of two continuous functions is also continuous. ■

Proposition 2.23 (Universal property). *Let $(X, \mathcal{T}_X)$, $(Z, \mathcal{T}_Z)$ be topological spaces and $A \subseteq X$ be with the subspace topology and $g : Z \rightarrow A$. Then g continuous iff $i \circ g$ continuous.*

$$\begin{array}{ccc} & & X \\ & \nearrow i \circ g & \uparrow i \\ Z & \xrightarrow{g} & A \end{array}$$

Proof. If g continuous $\implies (i \circ g)$ continuous by composition of continuous functions. Conversely, suppose we have $(i \circ g)$ continuous. We want that for any $\mathcal{U} \in \mathcal{T}_A$, $g^{-1}(\mathcal{U}) \in \mathcal{T}_Z$. $\mathcal{U} \in \mathcal{T}_A \implies \exists V \in \mathcal{T}$ such that $\mathcal{U} = V \cap A$. We have

$$g^{-1}(\mathcal{U}) = g^{-1}(V \cap A) = g^{-1}(i^{-1}(V)) = (i \circ g)^{-1}(V) \in \mathcal{T}_Z$$

by the continuity of $i \circ g$. So g is continuous. $\blacksquare$

Theorem 2.24

Given $(X, \mathcal{T})$, $A \subset X$, $\mathcal{T}_A$ is the only topology such that for any $(Z, \mathcal{T}_Z)$ and function $g : Z \rightarrow A$, g continuous iff $(i \circ g)$ is continuous.

Proof. Take a topology $\mathcal{T}'$ on A such that for any $(Z, \mathcal{T}_Z)$ and function $g : Z \rightarrow A$, g continuous iff $(i \circ g)$ is continuous.

(i) $\mathcal{T}_A \subseteq \mathcal{T}'$: Choose $(Z, \mathcal{T}_Z) = (A, \mathcal{T}')$ and $g = id$.

$$\begin{array}{ccc} & & (X, \mathcal{T}) \\ & \nearrow i \circ g & \uparrow i \\ (A, \mathcal{T}') & \xrightarrow{g=id} & (A, \mathcal{T}') \end{array}$$

Now $id = g$ is continuous $\implies i \circ g = i$ is continuous. So $\mathcal{T}_A \subseteq \mathcal{T}'$.

(ii) $\mathcal{T}' \subseteq \mathcal{T}_A$: Let $(Z, \mathcal{T}_Z) = (A, \mathcal{T}_A)$.

$$\begin{array}{ccc} & & (X, \mathcal{T}) \\ & \nearrow i \circ g & \uparrow i \\ (A, \mathcal{T}_A) & \xrightarrow{g=id} & (A, \mathcal{T}') \end{array}$$

$i \circ g = i : (A, \mathcal{T}_A) \rightarrow (X, \mathcal{T})$ but i is continuous. So $(i \circ g)$ is continuous $\implies g$ continuous, $g = id$. So $g^{-1}(\mathcal{U}) \in \mathcal{T}_A$ for any $\mathcal{U} \in \mathcal{T}'$. So $g^{-1}(\mathcal{U}) = \mathcal{U} \in \mathcal{T}_A$. So $\mathcal{T}' \subseteq \mathcal{T}_A$. $\blacksquare$

Lemma 2.25. *If $\mathcal{B}$ is a basis of $(X, \mathcal{T})$ and $A \subset X$, then $\mathcal{B}_A = \{\mathcal{U} \cap A : \mathcal{U} \in \mathcal{B}\}$ is a basis of $\mathcal{T}_A$.*

Proof. We need to show that if $\mathcal{U} \in \mathcal{T}_A$, then

$$\mathcal{U} = \bigcup_{i \in I} \mathcal{U}_i, \mathcal{U}_i \in \mathcal{B}_A$$

But if $\mathcal{U} \in \mathcal{T}_A$, then $\mathcal{U} = V \cap A$ for a $V \in \mathcal{T}$. So

$$V = \bigcup_{i \in I} V_i, V_i \in \mathcal{B}$$

Hence

$$\mathcal{U} = V \cap A = \left(\bigcup_{i \in I} V_i \right) \cap A = \bigcup_{i \in I} (V_i \cap A) = \bigcup_{i \in I} \mathcal{U}_i \in \mathcal{B}_A \quad \blacksquare$$

Given a metric space (X, d) and $A \subseteq X$, there are 2 canonical topologies:

- (i) Metric topology $\mathcal{T}_{(d_A)}$ on A
- (ii) Subspace topology $(\mathcal{T}_A)$

Proposition 2.26. $\mathcal{T}_{(d_A)} = (\mathcal{T}_A)$.

Proof. Exercise. (quite hard).

Products

Definition (Product topology). Let X and Y be topological spaces. Let $\mathcal{T}$ be the collection of all unions of subsets of $X \times Y$ of the form $\mathcal{U} \times V$ where $\mathcal{U}$ is open in X and V is open in Y . This topology on $X \times Y$ is called the *product topology*.

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Note: In $\mathbb{R}^2$, the black squares alone are not in the product topology; we also require the red squares for $X \times Y$:



Not every open set in the product topology is of the form $\mathcal{U} \times V$ with $\mathcal{U}$ open in X and V open in Y . But

$$\mathcal{B} = \{\mathcal{U} \times V : \mathcal{U} \text{ open in } X, V \text{ open in } Y\}$$

is a basis for the product topology. We will explore this on example sheets.

Exercise: The Euclidean topology on $\mathbb{R}^2$ (a) coincides with the product topology on $\mathbb{R} \times \mathbb{R}$ (b).

(a) is made up of unions of $B_\epsilon(x, y)$. Why is this open in the product topology? For each point in $B_\epsilon(x, y)$, we can surround it by another disk, but we can always inscribe a rectangle in this disk – this is the $\mathcal{U} \times V$ we use.

3 Hausdorff Spaces

Definition (Convergence). A sequence $(x_n)_{n \geq 1}$ of points in a topological space X converges to $x \in X$ written $x_n \rightarrow x$ if and only if for every open set $\mathcal{U}$ such that $x \in \mathcal{U}$, $\exists N$ such that whenever $n \geq N$ we have $x_n \in \mathcal{U}$.

Examples 3.1.

- For metric topologies this is the usual notion of convergence.
- X with the discrete topology, $x_n \rightarrow x$ if and only if $x_n = x$ for all $n \geq N$ (for some N).
- X with the indiscrete topology, $x_n \rightarrow x$ for all sequences $(x_n)_{n \geq 1}$ and all x .

Why don't sequences have unique limits here? Because open sets don't separate points.

Definition (Hausdorff). A topological space is called *Hausdorff* if and only if for all $x, y \in X$ with $x \neq y$, $\exists$ open sets $\mathcal{U}, V$ in X with $x \in \mathcal{U}$, $y \in V$ and $\mathcal{U} \cap V = \emptyset$.

Claim: Let X be a Hausdorff topological space. Then convergent sequences in X have a unique limit.

Proof. Let $(x_n)_{n \geq 1}$ be a sequence of points in X . Suppose that $x_n \rightarrow x$ and $x_n \rightarrow y$. We need to show that $x = y$. Suppose not. Then $x \neq y$ and $\exists$ open sets $\mathcal{U}, V$ such that $x \in \mathcal{U}$, $y \in V$ and $\mathcal{U} \cap V = \emptyset$. Since $x_n \rightarrow x$, $\exists N$ such that $x_n \in \mathcal{U}$ whenever $n \geq N$; since $x_n \rightarrow y$, $\exists M$ such that $x_n \in V$ whenever $n \geq M$. So for $n \geq \max(M, N)$, $x_n \in \mathcal{U}$ and $x_n \in V$ since $\mathcal{U} \cap V = \emptyset$. ■

Any metric topology is Hausdorff. (If (X, d) is a metric space and $x, y \in X$ satisfy $x \neq y$, then set $\epsilon = d(x, y)$, $\mathcal{U} = B_{\epsilon/2}(x)$, $V = B_{\epsilon/2}(y)$. Then $\mathcal{U}, V$ are open in X with $x \in \mathcal{U}$, $y \in V$ and $\mathcal{U} \cap V = \emptyset$.)

So any metrisable (there is a metric on X s.t. the topology is the metric topology) topological space X must be Hausdorff.

Example 3.2. Let X be an infinite set with the cofinite topology. Then X is not Hausdorff.

Open sets are all of the form $X \setminus \{\text{finitely many points}\}$. So if $x, y \in X$ and $\mathcal{U}, V$ are open sets with $x \in \mathcal{U}$, $y \in V$ we have $\mathcal{U} \cap V \neq \emptyset$. So $\mathcal{U} \cap V = X \setminus \{\text{finitely many points}\}$ and X is infinite, so $\mathcal{U} \cap V \neq \emptyset$.

Proposition 3.3.

- (i) A subspace of a Hausdorff space is Hausdorff.
- (ii) $X \times Y$ is Hausdorff $\iff X$ and Y are Hausdorff.
- (iii) If $f : X \rightarrow Y$ is a continuous injective map of topological spaces and Y is Hausdorff, then X is Hausdorff.

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(iv) If X and Y are homeomorphic, then X is Hausdorff $\iff Y$ is Hausdorff.

Proof.

(i) Let $a, b \in A$ be such that $a \neq b$. Then $a, b \in X$ and X is Hausdorff, so $\exists$ open subsets $\mathcal{U}, V$ of X such that $a \in \mathcal{U}$, $b \in V$ and $\mathcal{U} \cap V = \emptyset$. Set $\mathcal{U}' = \mathcal{U} \cap A$ and $V' = V \cap A$. Then $\mathcal{U}', V'$ are open in A (because A has the subspace topology) and $a \in \mathcal{U}'$ and $b \in V'$ and $\mathcal{U}' \cap V' = \emptyset$. So A is Hausdorff. $\square$

(ii) Suppose X, Y are both Hausdorff. Let (x_0, y_0) and (x_1, y_1) be distinct points of $X \times Y$. Then either $x_0 \neq x_1$ or $y_0 \neq y_1$. wlog exchanging X and Y if necessary, we may assume that $x_0 \neq x_1$. Then X is Hausdorff, so $\exists$ open sets $\mathcal{U}, V$ in X with $x_0 \in \mathcal{U}$, $x_1 \in V$ and $\mathcal{U} \cap V = \emptyset$. Set $\tilde{\mathcal{U}} = \mathcal{U} \times Y$ and $\tilde{V} = V \times Y$. Then $\tilde{\mathcal{U}}, \tilde{V}$ are open in $X \times Y$ (because $X \times Y$ has the product topology) and $(x_0, y_0) \in \tilde{\mathcal{U}}$ and $(x_1, y_1) \in \tilde{V}$ and $\tilde{\mathcal{U}} \cap \tilde{V} = \emptyset$ //

Suppose $X \times Y$ is Hausdorff. We will show that X is Hausdorff. Take $x_0, x_1 \in X$ with $x_0 \neq x_1$. Fix $y_0 \in Y$ and consider (x_0, y_0) and $(x_1, y_0) \in X \times Y$. These points are distinct and $X \times Y$ is Hausdorff $\implies \exists$ open sets $\tilde{\mathcal{U}}, \tilde{V}$ in $X \times Y$ such that $(x_0, y_0) \in \tilde{\mathcal{U}}$, $(x_1, y_0) \in \tilde{V}$ and $\tilde{\mathcal{U}} \cap \tilde{V} = \emptyset$. Because $\{\mathcal{U} \times V : \mathcal{U} \subset X \text{ open}, V \subset Y \text{ open}\}$ is a basis for the product topology, $\exists \mathcal{U}_1 \times V_1 \subseteq \tilde{\mathcal{U}}$ with $(x_0, y_0) \in \mathcal{U}_1 \times V_1$ and $\mathcal{U}_2 \times V_2 \subseteq \tilde{V}$ with $(x_1, y_0) \in \mathcal{U}_2 \times V_2$, and $\mathcal{U}_1, \mathcal{U}_2 \subseteq X$ are open, and $V_1, V_2 \subseteq Y$ are open.

We have $(\mathcal{U}_1 \times V_1) \cap (\mathcal{U}_2 \times V_2) = \emptyset$. I claim $\mathcal{U}_1 \cap \mathcal{U}_2 = \emptyset$. Suppose not. Then $\exists x' \in \mathcal{U}_1 \cap \mathcal{U}_2$ and $y_0 \in V_1 \cap V_2$ (because $y_0 \in V_1$ and $y_0 \in V_2$). So $(x', y_0) \in (\mathcal{U}_1 \times V_1) \cap (\mathcal{U}_2 \times V_2)$, $\mathbb{X}$, so $\mathcal{U}_1 \cap \mathcal{U}_2 = \emptyset$. But $x_0 \in \mathcal{U}_1$ and $x_1 \in \mathcal{U}_2$, with $\mathcal{U}_1, \mathcal{U}_2 \subseteq X$ open and $\mathcal{U}_1 \cap \mathcal{U}_2 = \emptyset$. So X is Hausdorff. $\square$

(iii) Let x_0, x_1 be distinct points of X . Then $f(x_0)$ and $f(x_1)$ distinct points of Y (because f is injective), so $\exists$ disjoint open sets $\mathcal{U}, V$ in Y with $f(x_0) \in \mathcal{U}$ and $f(x_1) \in V$ (because Y is Hausdorff). Set $\tilde{\mathcal{U}} = f^{-1}(\mathcal{U})$ and $\tilde{V} = f^{-1}(V)$. Then $\tilde{\mathcal{U}}, \tilde{V}$ are open in X (because f is continuous) and also $x_0 \in \tilde{\mathcal{U}}$ and $x_1 \in \tilde{V}$ and $\tilde{\mathcal{U}} \cap \tilde{V} = \emptyset$ (as if $\exists x' \in \tilde{\mathcal{U}} \cap \tilde{V}$ then $f(x') \in \mathcal{U} \cap V = \emptyset$, $\mathbb{X}$). Therefore X is Hausdorff. $\square$

(iv) Follows immediately from (iii) $\blacksquare$

Aside: When is a topological space metrisable? Being Hausdorff (can separate distinct points using disjoint open sets) is not enough. In metric spaces you can separate a point from a closed subset using disjoint open sets.

Definition. A topological space is regular iff for every closed subset $C \subseteq X$ and every $p \in X$ with $p \notin C$, $\exists$ open sets $\mathcal{U}, V \subseteq X$ such that $p \in \mathcal{U}$, $C \subseteq V$ and $\mathcal{U} \cap V = \emptyset$.

Metric spaces are regular. Hausdorff $\not\Rightarrow$ regularity.

4 Connected Spaces

Definition. A topological space X is *connected* iff $\nexists$ a continuous surjective map $f : X \rightarrow \{0, 1\}$ where $\{0, 1\}$ is given the discrete topology.

Equivalently X is connected iff $\nexists$ non-empty open sets $A, B \subseteq X$ such that $A \cup B = X$, $A \cap B = \emptyset$.

Why are these equivalent? We could take $A = f^{-1}(\{0\})$, $B = f^{-1}(\{1\})$ which are open and non-empty since f is surjective, with $A \cap B = \emptyset$.

Conversely define $f : X \rightarrow \{0, 1\}$ by

$$x \mapsto \begin{cases} 0 & x \in A \\ 1 & x \in B \end{cases}$$

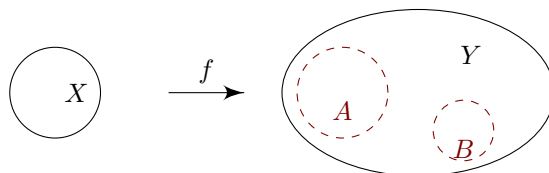
Examples 4.1.

- (i) Take $X = \{5, 7\}$ a subspace of $\mathbb{R}$. This is not connected; $A = \{5\}$, $B = \{7\}$. ($\{5\}$ is open since $\{5, 7\} \cap (4, 6) = \{5\}$)
- (ii) $\mathbb{Q}$ as a subspace of $\mathbb{R}$. *Connected or disconnected?* Disconnected: Take $A = (-\infty, \pi) \cap \mathbb{Q}$, $B = (\pi, \infty) \cap \mathbb{Q}$.
- (iii) X with the discrete topology is disconnected if $|X| > 1$. If $|X| > 1$, then take $A = \{x\}$, $B = X \setminus \{x\}$.
- (iv) X with the indiscrete topology is always connected: we can't choose non-empty open sets A, B with $A \cap B = \emptyset$.

Exercise: Show that the connected subsets of $\mathbb{R}$ are intervals. [I is an interval iff $x, y \in I \implies z \in I$ for all $x < z < y$.]

Proposition 4.2. Let X be a connected topological space, and let $f : X \rightarrow Y$ be continuous. Then $f(X)$ is connected.

Proof. Suppose not. Then choose non-empty A, B open subsets of $f(X)$ [with the subspace topology] such that $A \cap B = \emptyset$ and $A \cup B = f(X)$.



Set $\tilde{A} = f^{-1}(A)$, $\tilde{B} = f^{-1}(B)$. Then $\tilde{A}, \tilde{B}$ are open in X (since f continuous) and $\tilde{A} \cup \tilde{B} = X$ (as under f any $x \in X$ maps to either A or B) and $\tilde{A} \cap \tilde{B} = \emptyset$ (as $A \cap B = \emptyset$). So X is disconnected, $\mathbb{X}$. Thus $f(X)$ is connected. ■

Corollary 4.3. Suppose that topological spaces X, Y are homeomorphic. Then X is connected iff Y is connected.

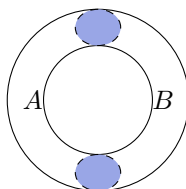
Proposition 4.4. Let $A_i, i \in I$ be connected subsets of a topological space X , such that $A_i \cap A_j \neq \emptyset$ for $i \neq j$. Then $\bigcup_{i \in I} A_i$ is connected.

Proof. Let $f : \bigcup_{i \in I} A_i \rightarrow \{0, 1\}$ be continuous. It suffices to show that f is not surjective. wlog not all the A_i 's are empty. On each A_i , the function f is constant. And these constants are the same on A_i and A_j for all i, j because $A_i \cap A_j \neq \emptyset$. So f is constant on $\bigcup_{i \in I} A_i$. Thus f is not surjective. ■

For a similar argument you can show that if B and $A_i, i \in I$ are connected subsets of X and $B \cap A_i \neq \emptyset$ for all i , then $B \cup \bigcup_{i \in I} A_i$ is connected.

Example 4.5. Suppose that A, B are connected subsets of X .

Does it follow that $A \cap B$ is connected? No! E.g. Two crescents intersecting is not connected:



By convention, $\emptyset$ is connected.

Lecture 16

Example 4.6. $X = \{x\}, \mathcal{T} = \{\emptyset, X\}$. Then X is connected.

Theorem 4.7

Let X and Y be topological spaces. Then $X \times Y$ is connected $\iff$ both X and Y are connected.

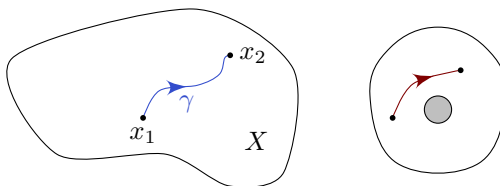
Proof. Suppose $X \times Y$ is connected. Let $p_X : X \times Y \rightarrow X$ and $p_Y : X \times Y \rightarrow Y$ be the projection maps; these are continuous. So $X = p_X(X \times Y)$ is connected and $Y = p_Y(X \times Y)$ is connected.

Last time: if B and $C_i, i \in I$ are connected and $B \cap C_i \neq \emptyset$ for all i , then $B \cup \bigcup_{i \in I} C_i$ is connected.

B is homeomorphic to X , hence connected, C_x is homeomorphic to Y , hence connected. $B \cap C_x = \{(x, y_0)\} \neq \emptyset$, so $B \cup \bigcup_{x \in X} C_x$ is connected. ■

4.1 Path Connectedness

Definition (Path). A *path* in a topological space X from $x_0 \in X$ to $x_1 \in X$ is $\gamma : [0, 1] \rightarrow X$ continuous such that $\gamma(0) = x_0, \gamma(1) = x_1$.

Examples 4.8.

Definition (Path connected). A topological space X is *path connected* iff any two points in X can be joined by a path in X .

Proposition 4.9. *If a topological space X is path-connected, then it is connected.*

Proof. Suppose X is disconnected. Then $\exists$ non-empty open sets $A, B \subseteq X$ such that $A \cup B = X$, $A \cap B = \emptyset$. Take $x_0 \in A, x_1 \in B$. Since X is path-connected, there is a path from x_0 to x_1 :

$$\gamma : [0, 1] \rightarrow X \text{ cts such that } \gamma(0) = x_0, \gamma(1) = x_1$$

Set $\tilde{A} = \gamma^{-1}(A), \tilde{B} = \gamma^{-1}(B)$. Then $\tilde{A}, \tilde{B}$ are open subsets of $[0, 1]$ (by continuity of γ), $\tilde{A} \cup \tilde{B} = [0, 1]$ and $\tilde{A} \cap \tilde{B} = \emptyset$, $\mathbb{X}$ since $[0, 1]$ is connected. So X is connected. ■

So path-connected $\implies$ connected, but connected $\not\Rightarrow$ path connected.

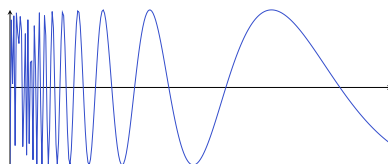
Exercise: Say that a topological space X is locally path connected iff every $x \in X$ has an open neighbourhood $\mathcal{U}$ such that $\mathcal{U}$ is path-connected.

- (i) Show that $\mathbb{R}^n$ is locally path-connected.
- (ii) If X is locally path connected, then X connected $\implies X$ path-connected.

In particular, for open subsets of $\mathbb{R}^n$, path-connected $\iff$ connected.

Example 4.10 (Topologists sine curve). (See Problem Sheet) X connected but not path connected : subspace of $\mathbb{R}^2$:

The graph of $y = \sin 1/x$ for $x > 0$ and y -axis from $y = -1$ to $y = 1$



Claim: $\mathbb{R}^2 \not\cong \mathbb{R}$

Proof. Suppose $\mathbb{R}^2$ is homeomorphic to $\mathbb{R}$. Then $\mathbb{R}^2 \setminus \{\text{one point}\}$ which is path-connected is homeomorphic to $\mathbb{R} \setminus \{\text{one point}\}$ which is disconnected. ■

Exercise: $S' = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\} \not\cong \mathbb{R}$.

5 Compactness

What does it mean for a topological space to be “small”?

Reasonable notion: X is small $\iff$ for every continuous function $f : X \rightarrow \mathbb{R}$, $f(X)$ is bounded. Motive: $[0, 1]$ is “small” in this sense. $(0, 1)$ is not small.

Definition (Compact). A subset A of a topological space X is *compact* if and only if every open cover of A admits a finite subcover.

Definition (Open cover). We say $\{\mathcal{U}_i : i \in I\}$ is an *open cover* of A iff each $\mathcal{U}_i$ is an open subset of X and $A \subseteq \bigcup_{i \in I} \mathcal{U}_i$. A *subcover* of $\{\mathcal{U}_i : i \in I\}$ is $\{\mathcal{U}_j : j \in J\}$ with $J \subseteq I$ such that $A \subseteq \bigcup_{j \in J} \mathcal{U}_j$. This is a *finite subcover* if $|J| < \infty$.

Example 5.1 (Non-example). $(0, 1)$ is not compact as a subset of $\mathbb{R}$.

Lecture 17

Proof. Consider $\mathcal{U}_i = (0, 1 - 1/i)$, $i \geq 2$. The $\mathcal{U}_i$ are open in $\mathbb{R}$ and $\bigcup_{i \geq 2} \mathcal{U}_i = (0, 1)$. We have

$$\mathcal{U}_2 \subseteq \mathcal{U}_3 \subseteq \mathcal{U}_4 \subseteq \dots$$

so if $\{\mathcal{U}_{i_1}, \dots, \mathcal{U}_{i_k}\}$ were a finite subcover then

$$\begin{aligned} (0, 1) &\subseteq \mathcal{U}_{i_1} \cup \dots \cup \mathcal{U}_{i_n} \\ &= \mathcal{U}_{\max(i_1, \dots, i_k)} = (0, \frac{i_k - 1}{i_k}) \end{aligned}$$

So $\{\mathcal{U}_i : i \geq 2\}$ has no finite subcover, $\mathbb{X}$. So $(0, 1)$ is not compact. $\blacksquare$

Note: $(0, 1)$ is not compact, but $\{(-1, 3)\}$ is an open cover of $(0, 1)$.

Example 5.2. $[0, 1]$ is compact as a subset of $\mathbb{R}$.

Proof. Let $\{\mathcal{U}_i : i \in I\}$ be an open cover of $[0, 1]$. We need to show that $\exists$ a finite subcover. Let

$$S = \{a \in [0, 1] : [0, a] \text{ can be covered by finitely many unions of } \mathcal{U}_i\}$$

S is non-empty because $0 \in S$. S is bounded above (by 1), so we can let $s = \sup S$. Then $s \in [0, 1]$. So $\exists j$ with $s \in \mathcal{U}_j$. Find $\epsilon > 0$ such that $(s - \epsilon, s + \epsilon) \subseteq \mathcal{U}_j$. Now $s = \sup S$, so $\exists a \in S$ with $a > s - \epsilon$. Then, if $s < 1$, we have that $[0, s + \frac{\epsilon}{2}]$ can be covered by finitely many of the $\mathcal{U}_i$'s, finitely many to cover $[0, a]$ by construction, plus $\mathcal{U}_j$ to cover $[a, s + \frac{\epsilon}{2}]$, $\mathbb{X}$. So $s = 1$. At this point I know that $[0, a]$ can be covered by finitely many $\mathcal{U}_i$ for all $a < 1$.

We need to show that $[0, 1]$ can be covered by finitely many $\mathcal{U}_i$. Now $1 \in \mathcal{U}_k$ for some k , so $1 - \epsilon \in \mathcal{U}_k$ for some $\epsilon > 0$. Then $[0, 1 - \epsilon]$ is covered by $\mathcal{U}_{i_1}, \dots, \mathcal{U}_{i_n}$ for some $i_1, \dots, i_n$, so $[0, 1]$ is covered by $\mathcal{U}_{i_1}, \dots, \mathcal{U}_{i_n}, \mathcal{U}_k$. $\blacksquare$

Proposition 5.3. *Any compact subset C of a metric space (X, d) is bounded.*

Proof. If $C = \emptyset$, there's nothing to prove. Else $\exists c \in C$. Set $\mathcal{U}_n = B_n(c)$. $\mathcal{U}_n$ is open and $C \subseteq \bigcup_{n \geq 1} \mathcal{U}_n$, so $\{\mathcal{U}_n : n \geq 1\}$ is an open cover of C . C is compact, so $\exists$ finite subcover $\mathcal{U}_{i_1}, \dots, \mathcal{U}_{i_k}$ say, but $\mathcal{U}_1 \subseteq \mathcal{U}_2 \subseteq \mathcal{U}_3 \subseteq \dots$, so $\mathcal{U}_{i_1} \cup \dots \cup \mathcal{U}_{i_k} = \mathcal{U}_{\max(i_1, \dots, i_k)}$. Set $N = \max(i_1, \dots, i_k)$. Then $C \subseteq B_N(c)$, so C is bounded. ■

X is a subspace of $\mathbb{R}^2$, $B_\epsilon(c) = \{x \in X : d(c, x) < \epsilon\}$. In particular, compact subsets of $\mathbb{R}^n$ are bounded.

Proposition 5.4. *Compact subsets of Hausdorff topological spaces are closed.*

Proof. Let X be a Hausdorff topological space. Let $C \subseteq X$ be compact. We need to show that $X \setminus C$ is open. Take $y \in X \setminus C$. We need to show $\exists$ an open set $\mathcal{U}$ with $y \in \mathcal{U}$ and $\mathcal{U} \subseteq X \setminus C$.

Given $c \in C$, $\exists \mathcal{U}_c, V_c$ with $y \in \mathcal{U}_c$, $c \in V_c$ and $\mathcal{U}_c \cap V_c = \emptyset$. Note that $\{V_c : c \in C\}$ is an open cover of C . So $\exists c_1, \dots, c_k \in C$ such that $V_{c_1}, \dots, V_{c_k}$ is an open cover of C . Let $\mathcal{U} = \mathcal{U}_{c_1} \cap \dots \cap \mathcal{U}_{c_k}$. This is open. Then $\mathcal{U} \subseteq \mathcal{U}_{c_i}$ for each i . So

$$\mathcal{U} \cap (V_{c_1} \cup \dots \cup V_{c_k}) = \emptyset$$

So $\mathcal{U} \cap C = \emptyset$ (because $C \subseteq V_{c_1} \cup \dots \cup V_{c_k}$). So $\mathcal{U}$ is an open set in X , with $y \in \mathcal{U}$ and $\mathcal{U} \subseteq X \setminus C$. So $X \setminus C$ is open ($\implies C$ is closed). ■

In particular, compact subsets of $\mathbb{R}^n$ are closed. (Later, $A \subseteq \mathbb{R}^n$ is compact $\iff A$ is closed and bounded.)

Proposition 5.5. *Let X, Y be topological spaces and $f : X \rightarrow Y$ a continuous map. If X is compact then $f(X)$ is compact.*

Proof. Left as an exercise.

So if X is a compact topological space and $f : X \rightarrow \mathbb{R}$ continuous, then $f(X)$ is bounded. In other words, compact topological spaces are “small” in the sense of the motivating discussion at the start of the section. Sharper result:

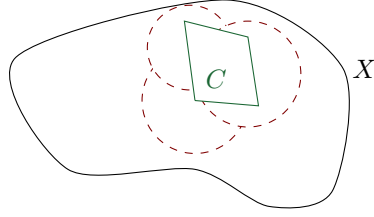
Lecture 18

Proposition 5.6. *Let X be a compact topological space and $f : X \rightarrow \mathbb{R}$ be continuous. Then f is bounded and attains its bounds.*

Proof. $f(X)$ is compact, and therefore bounded. So f is bounded. Notice also that $f(X)$ is closed (since $\mathbb{R}$ is Hausdorff). So $\inf f(X) \in f(X)$ and $\sup f(X) \in f(X)$. So $\exists x_+ \in X$ with $f(x) \leq f(x_+)$ for all $x \in X$, and $\exists x_- \in X$ with $f(x) \geq f(x_-)$ for all $x \in X$, and f attains its bounds. ■

Proposition 5.7. *A closed subset of a compact topological space is compact.*

Proof. Let X be a compact topological space, $C \subseteq X$ be closed.



Let $\{\mathcal{U}_i : i \in I\}$ be an open subcover of C . Need to show $\exists$ a finite subcover. Well, $\{\mathcal{U}_i : i \in I\} \cup \{X \setminus C\}$ is an open cover of X . So this has a finite subcover for X . This is either $\{\mathcal{U}_{i_1}, \dots, \mathcal{U}_{i_n}\}$ or $\{\mathcal{U}_{i_1}, \dots, \mathcal{U}_{i_n}\}, X \setminus C$. In either case $\{\mathcal{U}_{i_1}, \dots, \mathcal{U}_{i_n}\}$ covers C . ■

Note the closed condition is necessary, e.g. $[0, 1]$ is compact, $(0, 1)$ is not compact.

Theorem 5.8

Let X and Y be topological spaces. Then $X \times Y$ is compact iff both X and Y are compact.

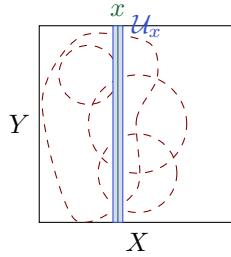
Proof. If $X \times Y$ is compact, then $X = p_X(X \times Y)$ is compact and $p_X : X \times Y \rightarrow X$ is continuous. Similarly Y is compact.

Suppose now that X and Y are compact, and that $\{W_i : i \in I\}$ is an open cover of $X \times Y$. We need to show $\exists$ finite subcover.

- (i) Step 1: Show that the ‘slice’ $\{x\} \times Y$ is covered by finitely many of the W_i . The slice is homeomorphic to Y , hence is compact and $\{W_i : i \in I\}$ is an open cover of the slice $\implies \exists W_{i_1}, \dots, W_{i_n}$ such that

$$\{x\} \times Y \subseteq W_{i_1} \cup \dots \cup W_{i_n}$$

- (ii) Step 2: $W_{i_1} \cup \dots \cup W_{i_n} \supseteq \mathcal{U}_x \times Y$ for an open set $\mathcal{U}_x \subseteq X$ containing x .



If $W_{i_j} = \mathcal{U}_j \times V_j$ for $\mathcal{U}_j \subseteq X$, $V_j \subseteq Y$ open then this is obvious:

$$\mathcal{U}_x = \mathcal{U}_{i_1} \cap \mathcal{U}_{i_2} \cap \dots \cap \mathcal{U}_{i_n}$$

Recall that $\{\mathcal{U} \times V : \mathcal{U} \subseteq X, V \subseteq Y \text{ open}\}$ is a basis for the product topology. Given (x, y) on my slice, we have $(x, y) \in W_{i_j}$ for some j , so $\exists$

open sets $Z_y \subseteq X, V_y \subseteq Y$ such that $(x, y) \subseteq Z_y \times V_y$. So $\{V_y : y \in Y\}$ is an open cover. This admits a finite subcover $V_{y_1}, \dots, V_{y_k}$, so $Y = V_{y_1} \cup \dots \cup V_{y_k}$. So I've covered by slice with finitely many rectangles:

$$Z_{y_1} \times V_{y_1}, \dots, Z_{y_k} \times V_{y_k}$$

and each rectangle is contained within one of the W_i 's. Thus $W_{i_1} \cup W_{i_n}$ contains

$$\underbrace{(Z_{y_1} \cap \dots \cap Z_{y_k})}_{\mathcal{U}_x} \times Y$$

(iii) Step 3: Do this for every $x \in X$. Then $\{\mathcal{U}_x : x \in X\}$ are an open cover of X . So $\exists$ finite subcover $\mathcal{U}_{x_1}, \dots, \mathcal{U}_{x_l}$. So

(1) $\mathcal{U}_{x_1} \times Y$ is covered by finitely many of the W_i : $W_{i_{1,1}}, \dots, W_{i_{1,m_1}}$

(2) $\mathcal{U}_{x_2} \times Y$ is covered by finitely many of the W_i : $W_{i_{2,1}}, \dots, W_{i_{2,m_2}}$

$\vdots$

(l) $\mathcal{U}_{x_l} \times Y$ is covered by finitely many of the W_i : $W_{i_{l,1}}, \dots, W_{i_{l,m_l}}$

So

$$\{W_{i_{a,b}} : 1 \leq a \leq l, 1 \leq b \leq n_a\}$$

covers all of $X \times Y$. ■

Corollary 5.9 (Heine-Borel Theorem). *Let A be a subset of $\mathbb{R}^n$. Then A is compact if and only if A is closed and bounded.* Lecture 19

Proof. $\implies$: $\mathbb{R}^n$ is a metric space, so compact subsets of $\mathbb{R}^n$ are bounded. $\mathbb{R}^n$ is Hausdorff, so compact subsets of $\mathbb{R}^n$ are closed.

$\impliedby$: Suppose that $A \subseteq \mathbb{R}^n$ is closed and bounded. Then $A \subseteq [a_1, b_1] \times \dots \times [a_n, b_n]$ for some $a_1, \dots, a_n, b_1, \dots, b_n$. But $[a, b]$ is compact, so $[a_1, b_1] \times \dots \times [a_n, b_n]$ is compact. And now A is a closed subset of a compact topological space hence is compact. ■

Aside: If $f : X \rightarrow Y$ is continuous and bijective, then it need *not* be a homeomorphism.

Claim: Let X be a compact topological space, Y be a Hausdorff topological space, $f : X \rightarrow Y$ be a continuous bijection. Then f is a homeomorphism.

Proof. We need to show that $f^{-1} : Y \rightarrow X$ is continuous, that is that whenever $\mathcal{U}$ is open in X , we have $(f^{-1})^{-1}(\mathcal{U})$ is open in Y . It is sufficient to prove that $f(V)$ is closed in Y whenever V is closed in X .

V is a closed subset of a compact topological space, so V is compact. Thus $f(V)$ is also compact (because f is continuous). But Y is Hausdorff, so $f(V)$ (being a compact subset of a Hausdorff space) is closed. ■

6 Sequential Compactness

Definition (Sequentially Compact). A metric space (X, d) is called *sequentially compact* iff every sequence $(x_n)_{n \geq 1}$ of points of X has a convergent subsequence. A subset A of a metric space (X, d) is called *sequentially compact* iff it is sequentially compact as a metric space with the subspace metric.

i.e. iff any sequence $(a_n)_{n \geq 1}$ of points of A has a subsequence which converges (in X) to a point of A .

E.g. closed bounded subsets of $\mathbb{R}^n$ are sequentially compact. $\mathbb{R}$ is not sequentially compact: $1, 2, 3, 4, 5, \dots$

Proposition 6.1. *A subset $A \subseteq \mathbb{R}$ is sequentially compact iff it is closed and bounded.*

Proof. Left as exercise.

Theorem 6.2

Let X be a metric space. Then X is sequentially compact iff X is compact.

Before we prove this, we prove the claim:

Claim: Let $(x_n)_{n \geq 1}$ be a sequence of points in a metric space (X, d) . Then $(x_n)_{n \geq 1}$ contains a subsequence converging to $x \in X$ iff for all $\epsilon > 0$, $B_\epsilon(x)$ contains infinitely many of the x_n .

Proof. $\implies$: If $(x_{n_k})_{k \geq 1}$ converges to x , then for all $\epsilon > 0$, $x_{n_k} \in B_\epsilon(x)$ whenever $k \geq K$ for some K . So $B_\epsilon(x)$ contains infinitely many of the x_n 's.

$\impliedby$: Construct a sequence $x_{n_1}, x_{n_2}, \dots$ as follows. Take $x_{n_k} \in B_{1/k}(x)$ such that $n_k > n_{k-1}$. ■

Proof of first half of theorem. $\impliedby$: Suppose not. Then $\exists$ a sequence $(x_n)_{n \geq 1}$ of points of X with no convergent subsequence. In particular, for any $x \in X$, there is no subsequence of $(x_n)_{n \geq 1}$ that converges to x . So $\exists \epsilon_x > 0$ such that $B_{\epsilon_x}(x)$ contains only finitely many of the x_n 's. Now $\{B_{\epsilon_x}(x) : x \in X\}$ is an open cover of X . But X is compact, so $\exists$ finite subcover $B_{\epsilon_{x_1}}(x_1), \dots, B_{\epsilon_{x_k}}(x_k)$. So $X = B_{\epsilon_{x_1}}(x_1) \cup \dots \cup B_{\epsilon_{x_k}}(x_k)$. So X is sequentially compact, ✕ ■

A new and useful concept before we prove $\implies$ direction of the theorem:

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Definition (Lebesgue number). Let A be a subset of a metric space (X, d) and $\{\mathcal{U}_i : i \in I\}$ a cover of A , we say that the cover has *Lebesgue number* $\epsilon > 0$ iff for all $a \in A$, $B_\epsilon(a) \subseteq \mathcal{U}_i$ for some $i \in I$.

If ϵ is a Lebesgue number for a cover, then ϵ' is a Lebesgue number whenever $0 < \epsilon' < \epsilon$.

Example 6.3. Consider $A = (0, 1) \subset \mathbb{R}$. $\mathcal{U}_1 = (-5, 2/3), \mathcal{U}_2 = (1/3, 17)$. Then $\epsilon = 1/3$ is not a Lebesgue number for $\{\mathcal{U}_1, \mathcal{U}_2\}$, since $B_{1/3}(a) \not\subseteq \mathcal{U}_i$ for $a \in [1/3, 2/3]$, but $\epsilon = 1/6$ is a Lebesgue number.

Claim: Any open cover of a sequentially compact metric space has a Lebesgue number.

Proof. Let X be a sequentially compact metric space. Suppose for contradiction that $\{\mathcal{U}_i : i \in I\}$ is an open cover of X with no Lebesgue number $\epsilon > 0$. Then $1/n$ is not a Lebesgue number, so $\exists x_n \in X$ such that $B_{1/n}(x_n) \not\subseteq \mathcal{U}_i$ for any $i \in I$. The sequence $(x_n)_{n \geq 1}$ has a convergent subsequence $(x_{n_k})_{k \geq 1}$ where $x_{n_k} \rightarrow x$ as $k \rightarrow \infty$.

But $\{\mathcal{U}_i : i \in I\}$ is an open cover of X , so $x \in \mathcal{U}_i$ for some $i \in I$. So $\exists \epsilon > 0$ such that $B_\epsilon(x) \subseteq \mathcal{U}_i$. For k sufficiently large, we have $d(x_{n_k}, x) < \epsilon/2$ and $1/n_k < \epsilon/2$, so $B_{1/n_k}(x_{n_k}) \subseteq B_\epsilon(x) \subseteq \mathcal{U}_i$, $\mathbb{X}$. So our assumption that the cover has no Lebesgue number must be false. ■

Definition (ϵ -net). Let X be a metric space and $\epsilon > 0$ be a real number. An ϵ -net in X is a subset $S \subseteq X$ such that $\bigcup_{s \in S} B_\epsilon(s) = X \iff \{B_\epsilon(s) : s \in S\}$ is an open cover for $X \iff$ every part of X is at most ϵ from a point of S .

Example 6.4. $\mathbb{Z}$ is a 1-net for $\mathbb{R}$. It is an ϵ -net for $\mathbb{R}$ whenever $\epsilon > 1/2$.

Claim: Let X be a sequentially compact metric space. Then $\forall \epsilon > 0, \exists$ finite subset $S \subseteq X$ that is an ϵ -net for X .

Proof. Suppose $\nexists$ a finite ϵ -net for X . Choose $x_1 \in X$. Then $\{x_1\}$ is not an ϵ -net. So $\exists x_2 \in X$ with $d(x_1, x_2) \geq \epsilon$. So $\{x_1, x_2\}$ is not an ϵ -net, so $\exists x_3$ such that $d(x_1, x_3) \geq \epsilon$ and $d(x_2, x_3) \geq \epsilon$. Continuing this, this gives a sequence $(x_n)_{n \geq 1}$ in X such that $d(x_i, x_j) \geq \epsilon$ whenever $i < j$. (So $d(x_i, x_j) \geq \epsilon$ whenever $i \neq j$). But $(x_n)_{n \geq 1}$ has a convergent subsequence, $x_{n_k} \rightarrow x$ say. So for k sufficiently large, $d(x_{n_k}, x) < \epsilon/2 \implies d(x_{n_k}, x_{n_k+1}) < \epsilon$, $\mathbb{X}$. Thus my original assumption that $\nexists$ a finite ϵ -net must have been false. ■

Lets finish the proof of Theorem 6.2: Sequentially compact metric spaces are compact.

Proof. Let X be a sequentially compact metric space and $\{\mathcal{U}_i : i \in I\}$ be an open cover of X . We need to show that there is a finite subcover. The cover $\{\mathcal{U}_i : i \in I\}$ has a Lebesgue number $\epsilon > 0$. So for any $x \in X$, $B_\epsilon(x) \subseteq \mathcal{U}_i$ for some $i \in I$. Also $\exists$ a finite ϵ -net for X , so X is covered by $B_\epsilon(x_1), \dots, B_\epsilon(x_n)$ for some $x_1, \dots, x_n \in X$. Then

$$B_\epsilon(x_1) \subseteq \{\mathcal{U}_{i_1}\}, \dots, B_\epsilon(x_n) \subseteq \{\mathcal{U}_{i_n}\}$$

So X is covered by $\mathcal{U}_{i_1}, \dots, \mathcal{U}_{i_n}$, so X is compact. ■

Remark. The Theorem is not true for topological spaces in general, though the examples of spaces that are compact but not sequentially compact are very complicated. But it's true for metric spaces.

Another application of Lebesgue number:

Definition (Uniformly continuous). Let $(X, d_X), (Y, d_Y)$ be metric spaces and $f : X \rightarrow Y$ be a map. We say that f is *uniformly continuous* iff given $\epsilon > 0$, $\exists \delta > 0$ such that whenever $d_X(x, x') < \delta$, we have $d_Y(f(x), f(x')) < \epsilon$.

This is stronger than f being continuous because the δ does not depend on x or x' ; it holds *uniformly* across to whole space X . So f uniformly continuous $\implies f$ continuous. Recall from M1P1: $f[a, b] \rightarrow \mathbb{R}$ is uniformly continuous $\iff f$ is continuous.

Proposition 6.5. Let $f : X \rightarrow Y$ be a continuous map between metric spaces and suppose that X is compact. Then f is uniformly continuous.

Proof. Fix $\epsilon > 0$. We need to show that $\exists \delta > 0$ such that whenever $d_X(x, x') < \delta$ we have $d_Y(f(x), f(x')) < \epsilon$. For a fixed x , $\exists \delta_x > 0$ such that whenever $d_X(x, x') < \delta_x$, we have $d_Y(f(x), f(x')) < \epsilon/2$. But $\{B_{\delta_x}(x) : x \in X\}$ is an open cover of X . So it has a Lebesgue number, $\delta > 0$. So whenever $z, z' \in X$ are such that $d(z, z') < \delta$, we have $z, z' \in B_\delta(x)$ and $B_\delta(z) \subseteq B_{\delta_x}(x)$ for some $x \in X$. So

$$\begin{cases} d_X(z, x) < \delta_x \\ d_X(z', x) < \delta_x \end{cases} \implies \begin{cases} d_Y(f(z), f(x)) < \epsilon/2 \\ d_Y(f(z'), f(x)) < \epsilon/2 \end{cases}$$

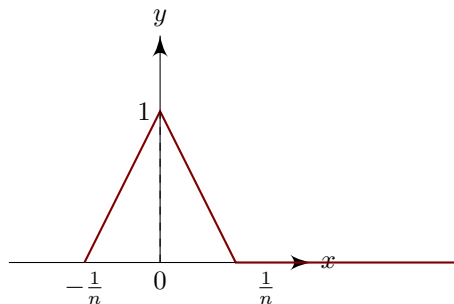
And so $d_Y(f(z), f(z')) < \epsilon$. ■

6.1 Convergence of Functions

Definition (Pointwise convergence). Let S be a set and $f_n : S \rightarrow \mathbb{R}$ be functions, $n \in \mathbb{N}$ and let $f : S \rightarrow \mathbb{R}$ be a function. We say that $f_n \rightarrow f$ *pointwise* iff $f_n(s) \rightarrow f(s)$ for all $s \in S$.

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Example 6.6. Take $S = [-1, 1]$. Take f_n as



Then $f_n \rightarrow f$ pointwise, where

$$f = \begin{cases} 0 & x \neq 0 \\ 1 & x = 0 \end{cases}$$

Definition. We say that $f_n \rightarrow f$ *uniformly* if for all $\epsilon > 0$, $\exists N$ such that whenever $n \geq N$, we have $|f_n(s) - f(s)| < \epsilon$ for all $s \in S$.

Proposition 6.7. Let $f_n : X \rightarrow Y$ be continuous functions between metric spaces X, Y and suppose that $f_n \rightarrow f$ uniformly for some $f : X \rightarrow Y$. Then f is continuous.

Proof. This is an $\epsilon/3$ argument. Fix $\epsilon > 0$. Now $f_n \rightarrow f$ uniformly, so $\exists N$ such that for all $n \geq N$, we have $d_Y(f_n(x), f(x)) < \epsilon/3$ for all $x \in X$. Also f_N is continuous, so for all $x \in X$, $\exists \delta$ such that $d_Y(f_N(x), f_N(x')) < \epsilon/3$ whenever $d_X(x, x') < \delta$. Thus if $d_X(x, x') < \delta$, we have

$$\begin{aligned} d_Y(f(x), f(x')) &\leq d_Y(f(x), f_N(x)) + d_Y(f_N(x), f_N(x')) + d_Y(f_N(x'), f(x')) \\ &< \epsilon/3 + \epsilon/3 + \epsilon/3 \\ &= \epsilon \end{aligned}$$

So f is continuous at x , which was arbitrary, so f is continuous. ■

Rephrased: Let X, Y be metric spaces. Define

$$B(X, Y) := \{f : X \rightarrow Y : f \text{ is bounded.}\}$$

and define $d(f, g) := \sup_{x \in X} d_Y(f(x), g(x))$. This is a metric on $B(X, Y)$. We can check that usual conditions hold.

What does convergence in this metric mean? Suppose $f_n \in B(X, Y)$ and $f \in B(X, Y)$ and that $f_n \rightarrow f$ in this metric. Well,

$$\begin{aligned} f_n \rightarrow f &\iff d(f_n, f) \rightarrow 0 \text{ as } n \rightarrow \infty \\ &\iff \sup_{x \in X} d_Y(f_n(x), f(x)) \rightarrow 0 \text{ as } n \rightarrow \infty \\ &\iff \forall \epsilon > 0, \exists N : n \geq N \implies \sup_{x \in X} d_Y(f_n(x), f(x)) < \epsilon \\ &\iff \forall \epsilon > 0, \exists N : n \geq N \implies d_Y(f_n(x), f(x)) < \epsilon \forall x \in X \\ &\iff f_n \rightarrow f \text{ uniformly} \end{aligned}$$

We just said that the uniform limit of continuous functions is continuous. Consider,

$$C^{bdd}(X, Y) := \{f : X \rightarrow Y : f \text{ continuous and bounded}\}$$

Then $C^{bdd}(X, Y) \subset B(X, Y)$. What we just showed is that $C^{bdd}(X, Y)$ is *closed* in $B(X, Y)$ [because if $f \in B(X, Y)$ is a point of closure of $C^{bdd}(X, Y)$, then $\exists (f_n)_{n \geq 1}$ in $C^{bdd}(X, Y)$ with $f_n \rightarrow f$. So f is the uniform limit of continuous functions, so $f \in C^{bdd}(X, Y)$.]

7 Complete Metric Spaces

Definition (Completeness). A metric space is *complete* iff every Cauchy sequence in X converges (to a point of X).

Examples 7.1. $\mathbb{R}$ is complete, $[0, 1]$ is complete. $\mathbb{N}$ is complete (as a subspace of $\mathbb{R}$). $\mathbb{Q} \subset \mathbb{R}$ is *not* complete. $(0, 1)$ is not complete, since $x_n = 1 - 1/n \rightarrow 1$ but $1 \notin (0, 1)$. Notice $(0, 1) \cong \mathbb{R}$, but the LHS is not complete, and the RHS is. So completeness is not a topological property.

Recap: A Cauchy sequence $(x_n)_{n \geq 1}$ in a metric space (X, d) satisfies: for all $\epsilon > 0, \exists N$ such that $d(x_n, x_m) < \epsilon$ whenever $n, m \geq N$. Lecture 22

A sequence $(x_n)_{n \geq 1}$ converges in X always $\implies$ the sequence $(x_n)_{n \geq 1}$ is Cauchy ($\Leftarrow$ in complete metric spaces.)

Proof. Suppose that $(x_n)_{n \geq 1}$ converges to $x \in X$. Given $\epsilon > 0, \exists N$ such that $d(x_n, x) < \epsilon/2$ whenever $n \geq N$. So if $n, m \geq N$, we have

$$\begin{aligned} d(x_n, x_m) &\leq d(x_n, x) + d(x, x_m) \\ &< \epsilon/2 + \epsilon/2 = \epsilon \end{aligned}$$

So $(x_n)_{n \geq 1}$ is Cauchy. ■

Completeness is *not* a topological property: $(0, 1) \cong \mathbb{R}$, however $(0, 1)$ is not complete and $\mathbb{R}$ is complete. But...

Proposition 7.2. Let X, Y be metric spaces and let $f : X \rightarrow Y$ be a bijection such that f and f^{-1} are uniformly continuous. Then X is complete $\iff Y$ is complete.

Proof. Done in assessed coursework #2.

Recall that two metrics d_1, d_2 on a set X are called *Lipschitz-equivalent* iff $\exists h, k > 0$ constants such that

$$hd_2(x, y) \leq d_1(x, y) \leq kd_2(x, y)$$

for all $x, y \in X$.

Lipschitz-equivalent metrics define the same topology on X .

Proof. The identity map $(X, d_1) \rightarrow (X, d_2)$ is uniformly continuous and has uniformly continuous inverse. ■

7.1 Properties of complete metric spaces

Proposition 7.3. A complete metric subspace Y of a metric space X is closed in X .

Proof. We need to show that $\bar{Y} \subseteq Y$. Let $x \in \bar{Y}$. Then $\exists$ sequence $(y_n)_{n \geq 1}$ of points of Y with $y_n \rightarrow x$. Then $(y_n)_{n \geq 1}$ is convergent in X , hence it is a Cauchy sequence. And Y is complete, so $y_n \rightarrow y$ for some $y \in Y$. But convergent sequences have unique limits (because X is Hausdorff), so $y = x$. Thus $x \in Y$. ■

Similarly:

Proof. We need to show that every Cauchy sequence in Y converges to a point of Y . Let $(y_n)_{n \geq 1}$ be a Cauchy sequence in Y . Since X is complete, we know that $y_n \rightarrow x$ for some $x \in X$. And then x is a point of closure of Y . So $x \in \bar{Y} = Y$. Thus $(y_n)_{n \geq 1}$ converges to a point of Y . ■

Proposition 7.4. *Compact metric spaces are complete.*

Proof. Let X be a compact metric space and let $(x_n)_{n \geq 1}$ be a Cauchy sequence in X . We need to show that $(x_n)_{n \geq 1}$ converges. X is sequentially compact, so $\exists$ subsequence $(x_{n_k})_{k \geq 1}$ such that $x_{n_k} \rightarrow x$ as $k \rightarrow \infty$. So it suffices to prove:

Claim: Let $(x_n)_{n \geq 1}$ be a Cauchy sequence in a metric space X and suppose that a subsequence $(x_{n_k})_{k \geq 1}$ converges to x as $k \rightarrow \infty$. Then $x_n \rightarrow x$ as $n \rightarrow \infty$.

Proof of Claim. Fix $\epsilon > 0$. We need to show $\exists N$ such that $d(x_n, x) < \epsilon$ whenever $n \geq N$. Now $x_{n_k} \rightarrow x$ as $k \rightarrow \infty$, so $\exists K$ such that $d(x_{n_k}, x) < \epsilon/2$ whenever $k \geq K$. And $(x_n)_{n \geq 1}$ is Cauchy, so $\exists N'$ such that $d(x_n, x_m) < \epsilon/2$ whenever $n, m \geq N'$. Set $N = \max(n_K, N')$. Then for $n \geq N$ we have

$$\begin{aligned} d(x_n, x) &\leq d(x_n, x_{n_K}) + d(x_{n_K}, x) \\ &< \epsilon/2 + \epsilon/2 = \epsilon \end{aligned}$$

So $x_n \rightarrow x$ as claimed. This completes the proof that compact metric spaces are complete. ■

Proposition 7.5. *Let X, Y be metric spaces. Then $X \times Y$ complete $\iff$ both X and Y are complete.*

Proof. Left as an exercise. (Use d_∞).

Corollary 7.6. $\mathbb{R}^n$ is complete.

7.2 Contraction Mapping Theorem

Definition (Contraction). Let (X, d) be a metric space. A map $f : X \rightarrow X$ is called a *contraction* if and only if $\exists K < 1$ such that $d(f(x), f(y)) \leq Kd(x, y)$ for all $x, y \in X$.

Lecture 23

Note that contractions are automatically uniformly continuous.

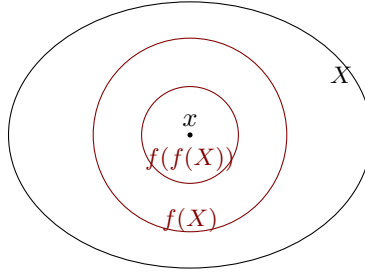
[Aside: $d(f(x), f(y)) < d(x, y) \forall x, y \in X \implies f$ is continuous.]

Definition (Fixed point). A *fixed point* of $f : X \rightarrow X$ is $x \in X$ such that $f(x) = x$.

Theorem 7.7

If X is a complete metric space and $f : X \rightarrow X$ is a contraction, then $\exists$ a unique fixed point for f in X .

Proof. Intuition:



I claim that this x is a fixed point. In other words, $X \supset f(X) \supset f^2(X) \cdots \supset f^n(X) \supset \dots$,

$$\bigcup_{n=0}^{\infty} f^{(n)}(X) = \{x\} \text{ and } f(x) = x$$

Strategy: Pick x_0 as an arbitrary point of X and set $x_n = f(x_{n-1})$, $n \geq 1$ to get a sequence $(x_n)_{n \geq 0}$ in X . Show that $x = \lim_{n \rightarrow \infty} x_n$ exists and satisfies $f(x) = x$.

Claim: $(x_n)_{n \geq 0}$ is a Cauchy sequence.

Proof of claim:

$$\begin{aligned} d(x_0, x_1) &= \text{something} \\ d(x_1, x_2) &= d(f(x_0), f(x_1)) \leq Kd(x_0, x_1) \\ d(x_2, x_3) &= d(f(x_1), f(x_2)) \leq Kd(x_1, x_2) \leq K^2d(x_0, x_1) \\ &\vdots \\ d(x_n, x_{n+1}) &\leq K^n d(x_0, x_1) \end{aligned}$$

Suppose $m \geq n$, then

$$\begin{aligned} d(x_n, x_m) &\leq d(x_n, x_{n+1}) + d(x_{n+1}, x_{n+2}) + \cdots + d(x_{m-1}, x_m) \\ &\leq (K^n + K^{n+1} + \cdots + K^m) d(x_0, x_1) \\ &\leq (K^n + K^{n+1} + \cdots + K^m + \cdots) d(x_0, x_1) \\ &= \frac{K^n}{1-K} d(x_0, x_1) \rightarrow 0 \text{ as } n \rightarrow \infty \end{aligned}$$

So $(x_n)_{n \geq 0}$ is a Cauchy sequence. Thus $x_n \rightarrow x$ for some $x \in X$. (because X is complete). $\square$

Claim: $f(x) = x$.

Proof of claim: $x_n \rightarrow x$, so $f(x_n) \rightarrow f(x)$ [because f is continuous] and $x_{n+1} \rightarrow x$, so $f(x) = x$. $\square$

Uniqueness: Suppose $x \in X$ and $y \in X$ are fixed point for f . We need to show that $x = y$. But

$$d(x, y) = d(f(x), f(y)) \leq Kd(x, y)$$

and $K < 1$, so $d(x, y) = 0$. Thus $x = y$. $\blacksquare$

Example 7.8. $X = [0, \frac{1}{2}]$, $f : X \rightarrow X$, where $x \mapsto x^2$. This is a contraction, so $\exists$ a unique fixed point in $[0, \frac{1}{2}]$, i.e. a unique solution to $x^2 = x$ in $[0, \frac{1}{2}]$.

Completeness is essential; the same f but over $X = (0, \frac{1}{2})$ does not have a fixed point even though f is a contraction.

Example 7.9.* Suppose that

$$D = [x_n - a, x_n + a] \times [y_n - b, y_n + b]$$

and $f : D \rightarrow \mathbb{R}$ is continuous and satisfies

$$|f(x, y_1) - f(x, y_2)| \leq K|y_1 - y_2|$$

for all (x, y_1) and $(x, y_2) \in D$, for some $K > 0$. Let M be any upper bound for $|f(x, y)|$ on D and fix $c > 0$ such that $c < \min(a, b/M, 1/K)$. Then on $I = [x_0 - c, x_0 + c]$, there exists a unique solution $y(x)$ to

$$\frac{dy}{dx} = f(x, y); \quad y(x_0) = y_0 \quad (*)$$

Proof. $(*)$ holds if and only if $x \mapsto y(x)$ is a continuous solution to

$$y(x) = y_0 + \int_{t=x_0}^{t=x} f(t, y(t)) \, dt$$

If $\frac{dy}{dx} = f(x, y)$ and $y(x_0) = y_0$, then

$$\int_{t=x_0}^{t=x} f(t, y(t)) \, dt = \int_{t=x_0}^{t=x} \frac{dy}{dt} \, dt = y(x) - y(x_0)$$

On the other hand, if

$$y(x) = y_0 + \int_{t=x_0}^{t=x} f(t, y(t)) \, dt$$

then $\frac{dy}{dx} = f(x, y(x))$ by the fundamental theorem of calculus.

Let $X \subseteq C(I)$ consist of $f \in C(I)$ such that

$$|f(x) - y_0| \leq b, \forall x \in I$$

X is closed in $C(I)$ and $C(I)$ is complete. So X is complete. Consider $F : X \rightarrow X$ with $y \mapsto F(y)$, where

$$F(y)(x) = y_0 + \int_{t=x_0}^{t=x} f(t, y(t)) \, dt$$

If y is continuous then $F(y)$ is continuous (so $F : X \rightarrow C(I)$).

Claim: If $|y(x) - y_0| \leq b$ for all $x \in I$, then $|F(y)(x) - y_0| \leq b$ for all $x \in I$, because

$$\begin{aligned} |F(y)(x) - y_0| &= \left| \int_{t=x_0}^{t=x} f(t, y(t)) \, dt \right| \\ &\leq (x - x_0)M \leq cM \leq b \end{aligned}$$

(so $F : X \rightarrow X$).

Claim: F is a contraction.

Proof. Let $y_0, y_1 \in X$. Then

$$\begin{aligned} d(F(y_0), F(y_1)) &= \sup_{x \in [x_0 - c, x_0 + c]} |F(y_0)(x) - F(y_1)(x)| \\ &\leq cKd(y_0, y_1) \end{aligned}$$

with $cK < 1$. Hence F has a unique fixed point, which is a unique continuous solution of (*). ■

Final Remarks

Aside: Let X be a topological space and $A \subseteq X$ a subset. When is A compact as a subset of X ? Lecture 24

Consider A with the subspace topology. When is A compact? I claim these two questions are the same. For

A is compact as a subset of X

$\iff$ every open cover of A has finite subcover

$\iff$ whenever $\{\mathcal{U}_i, i \in I\}$ open in X with $A \subseteq \bigcup_i \mathcal{U}_i$, $\exists \mathcal{U}_{i_1}, \dots, \mathcal{U}_{i_n}$ with $A \subseteq \bigcup_{j=1}^n \mathcal{U}_{i_j}$

When is A (with subspace topology) a compact topological space?

A is compact $\iff$ whenever $\{V_i : i \in I\}$ open in A with subspace topology, covering A , $\exists V_{i_1}, \dots, V_{i_n}$ s.t. $A \subseteq \bigcup_{j=1}^n V_{i_j}$.

But V_i is open in the subspace topology of A iff $V_i = \mathcal{U}_i \cap A$ where $\mathcal{U}_i$ open in X . Choose $\mathcal{U}_i$ open in X such that $V_i = \mathcal{U}_i \cap A$. Then $\{\mathcal{U}_i : i \in I\}$ is an open cover of A in X iff $\{V_i : i \in I\}$ is an open cover for A in the subspace topology.

Aside:

Definition (Density). A subspace A of a topological space X is *dense* if and only if $\overline{A} = X$.

Examples 7.10.

- (i) $\mathbb{Q}$ is dense in $\mathbb{R}$. $\mathbb{R} \setminus \mathbb{Q}$ is also dense in $\mathbb{R}$.
- (ii) Let X be a non-empty set with the discrete topology. Then $A \subseteq X$ is dense $\iff A = X$, since all sets are closed.
- (iii) Let X be a non-empty set with the indiscrete topology. Then $A \subseteq X$ is dense $\iff A \neq \emptyset$.

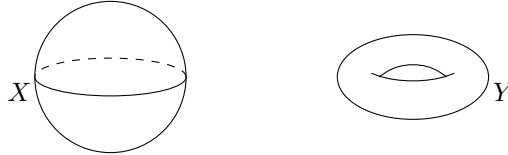


"You left this on our ship."

8 The Fundamental Group

Question: Suppose X is the two-dimensional sphere, i.e.

$$X = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$$

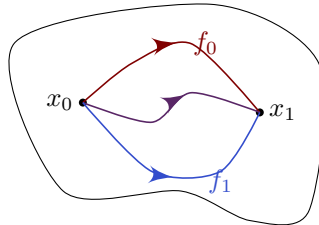


and suppose Y is the two-dimensional torus, $S^1 \times S_1$ where $S_1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$. Are these two homeomorphic?

We'd like to think not, since Y has a hole in it; there'd surely have to be a discontinuity in any mappings. We're going to build an algebraic structure out of these 'loops' to be able to formalise this.

Recall a path in a topological space X is $\gamma : [0, 1] \rightarrow X$ continuous. The end points of the path are $x_0 = \gamma(0), x_1 = \gamma(1)$. The path is a loop if $x_0 = x_1$.

Definition (Homotopy). A *homotopy* between paths $f_0 : [0, 1] \rightarrow X$ and $f_1 : [0, 1] \rightarrow X$, each of which is a path from $x_0 \in X$ to $x_1 \in X$, is a family $f_t : [0, 1] \rightarrow X$ of paths from x_0 to x_1 such that the map $F : [0, 1] \times [0, 1] \rightarrow X$ mapping $(s, t) \mapsto f_t(s)$ is continuous.



Definition (Homotopic Paths). We say that paths f_0, f_1 are *homotopic* iff $\exists$ a homotopy from f_0 to f_1 .

Homotopy is an equivalence relation:

- Reflexive: $f \sim f$ – given a path $f : [0, 1] \rightarrow X$ we need a homotopy from f to f . Taking $f_t(s) = f(s)$ works.
- Symmetry: If $f \sim g$ then $g \sim f$; Let f, g be paths from x_0 to x_1 in X that are homotopic. So $\exists$ paths $f_t : [0, 1] \rightarrow X$ from x_0 to x_1 where $t \in [0, 1]$ such that $f_0 = f, f_1 = g$ and $[0, 1] \times [0, 1] \rightarrow X$ where $(s, t) \mapsto f_t(s)$ is continuous. Set $g_t = f_{1-t}(s)$. We need to show that $(s, t) \mapsto f_{1-t}(s)$ is continuous: Composing the two functions is the identity, which is continuous, so it must be continuous as well.
- Transitive: Left as *exercise*.

Lemma 8.1 (Pasting). *Let X be a topological space and let A, B be closed subsets of X such that $X = A \cup B$. Suppose that $f : A \rightarrow Y$ and $g : B \rightarrow Y$ are continuous functions such that $f(x) = g(x)$ for all $x \in A \cap B$. Then $h : X \rightarrow Y$ mapping*

$$x \mapsto \begin{cases} f(x) & x \in A \\ g(x) & x \in B \end{cases}$$

is continuous.

Proof. It suffices to show that whenever $V \subseteq Y$ is closed, $h^{-1}(V)$ is closed in X . But $h^{-1}(V) = f^{-1}(V) \cup g^{-1}(V)$ and $f^{-1}(V)$ is closed in A , hence closed in X . Similarly $g^{-1}(V)$ is closed in X . So $h^{-1}(V)$ is the union of two closed sets and so is closed. ■

We could take A, B to be open instead and the result would still hold (*exercise.*) but we need some condition on A, B for the statement to hold, e.g.

Example 8.2. Let $X = \mathbb{R}, A = \mathbb{Q}, B = \mathbb{R} \setminus \mathbb{Q}$ and take $f : A \rightarrow \mathbb{R}$ with $x \mapsto 1$ and $g : B \rightarrow \mathbb{R}$ with $x \mapsto 0$. These are definitely continuous, but $h : \mathbb{R} \rightarrow \mathbb{R}$ sending

$$x \mapsto \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

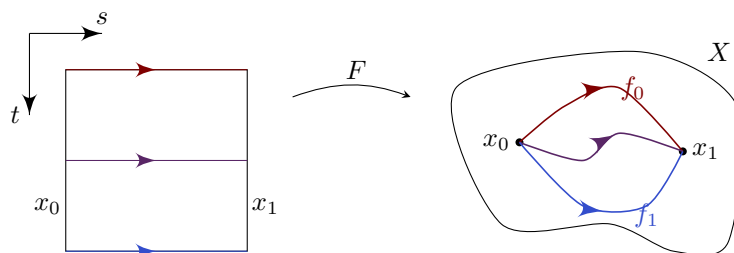
is not continuous.

Notation. $I = [0, 1]$.

Recall:

A *path* in a topological space X from $x_0 \in X$ to $x_1 \in X$ is $f : I \rightarrow X$ continuous such that $f(0) = x_0, f(1) = x_1$.

Given two paths, $f_0 : I \rightarrow X, f_1 : I \rightarrow X$, from $x_0 \in X$, a *homotopy* is a family of paths $f_t : I \rightarrow X, 0 \leq t \leq 1$ from x_0 to x_1 such that $I \times I \rightarrow X$ mapping $(x, t) \mapsto f_t(s)$.

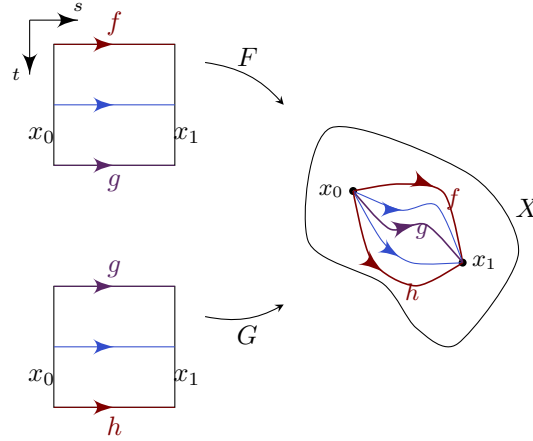


f is homotopic to g , “ $f \sim g$ ” $\iff \exists$ a homotopy from f to g .

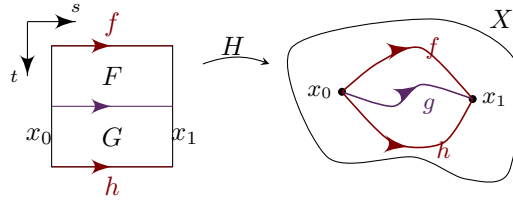
Claim: $\sim$ is an equivalence relation.

Proof. Last time we checked reflexive and symmetric. Now transitive:

Let $f, g, h : I \rightarrow X$ be paths from $x_0 \in X$ to $x_1 \in X$. Let $F : I \times I \rightarrow X$ be a homotopy from g to f and let $G : I \times I \rightarrow X$ be a homotopy from g to h . We need to find a homotopy from f to h .



We seek:



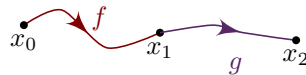
i.e. $H : I \times I \rightarrow X$ mapping

$$(s, t) \mapsto \begin{cases} F(s, 2t) & 0 \leq t \leq \frac{1}{2} \\ G(s, 2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

H is continuous because of the pasting lemma. ■

The composition of loops

Given a path $f : I \rightarrow X$ from x_0 to x_1 and a path $g : I \rightarrow X$ from x_1 to x_2 , I can compose them to get a path $f * g$ from x_0 to x_2 .



Then $f * g : I \rightarrow X$ with

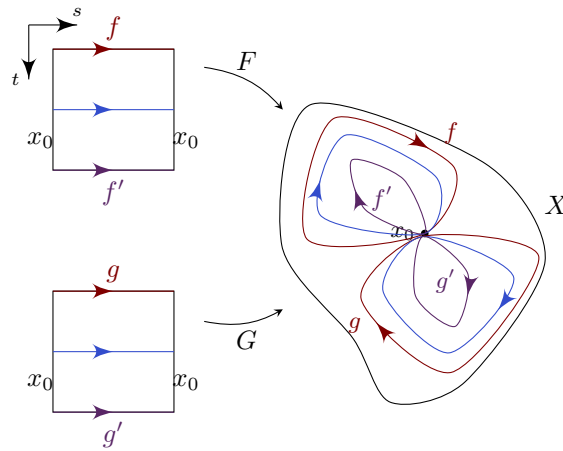
$$t \mapsto \begin{cases} f(2t) & 0 \leq t \leq \frac{1}{2} \\ g(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

$f * g$ is continuous by the pasting lemma.

Fix $x_0 \in X$ (the *basepoint*) and consider new loops based at x_0 , i.e. paths $f : I \rightarrow X$ such that $f(0) = f(1) = x_0$. Then the composition of paths defines a “multiplication” operation on this set of loops.

Claim: Let f, f', g, g' be loops based at $x_0 \in X$. Suppose that $f \sim f'$ and $g \sim g'$. Then $f * g \sim f' * g'$. (so composition of paths respects the equivalence relation of homotopy).

Proof. Let $F : I \times I \rightarrow X$ be a homotopy from f to f' . Let $G : I \times I \rightarrow X$ be a homotopy from g to g' . We need to construct a homotopy from $f * g$ to $f' * g'$.



So consider

$$H(s, t) = \begin{cases} F(2s, t) & 0 \leq s \leq \frac{1}{2} \\ G(2s - 1, t) & \frac{1}{2} \leq s \leq 1 \end{cases}$$

This is continuous (pasting lemma) and is a homotopy from $f * g$ to $f' * g'$. ■

Definition (The Fundamental Group). The *fundamental group* is

$$\pi_1(X, x_0) = \{[\gamma] : \gamma \text{ is a loop based at } x_0\}$$

with the product

$$[\gamma] * [\gamma'] = [\gamma * \gamma']$$

Claim: $\pi_1(X, x_0)$ is a group.

Lecture 26

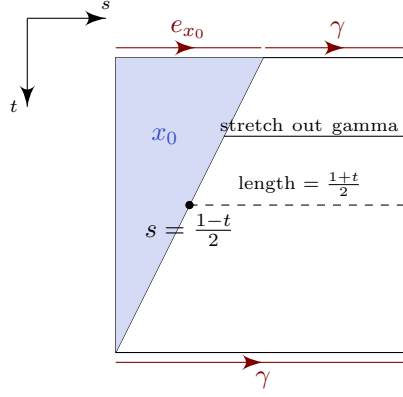
Proof. We need

- (i) **IDENTITY:** Set $e_{x_0} \in \pi_1(X, x_0)$ to be the constant loop at x_0 , i.e. $e_{x_0}(t) = x_0$ for all $t \in [0, 1]$. I claim that for all $[\gamma] \in \pi_1(X, x_0)$,

$$[e_{x_0}] * [\gamma] = [\gamma] \text{ and } [\gamma] * [e_{x_0}] = [\gamma]$$

Indeed,

$$e_{x_0} * \gamma(t) = \begin{cases} x_0 & 0 \leq t \leq \frac{1}{2} \\ \gamma(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

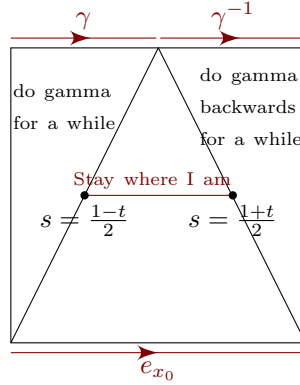


So take the homotopy

$$F(s, t) = \begin{cases} x_0 & 0 \leq s \leq \frac{1}{2}(1-t) \\ \gamma\left(\frac{2}{t+1}\left(s - \frac{1}{2}(1-t)\right)\right) & \frac{1}{2}(1-t) \leq s \leq 1 \end{cases}$$

Right inverse is similar. So e_{x_0} is the identity element in $\pi_1(X, x_0)$.

- (ii) INVERSES. Aside: Given a path γ from x_0 to x_1 , define $\gamma^{-1} : I \rightarrow X$, mapping $t \mapsto \gamma(1-t)$. This is a path from x_1 to x_0 . So we need to show that $\gamma * \gamma^{-1} = e_{x_0}$ and $\gamma^{-1} * \gamma = e_{x_0}$. We need a homotopy from $\gamma * \gamma^{-1}$ to the constant loop e_{x_0} :



So define

$$F(s, t) = \begin{cases} \gamma(2s) & 0 \leq s \leq \frac{1}{2} - \frac{1}{2}t \\ \gamma(1-t) & \frac{1}{2} - \frac{1}{2}t \leq s \leq \frac{1}{2} + \frac{1}{2}t \\ \gamma^{-1}(2s - 1) & \frac{1}{2} + \frac{1}{2}t \leq s \leq 1 \end{cases}$$

We need F to be continuous, $F(s, 0) = \gamma * \gamma^{-1}(s)$, and $F(s, 1) = e_{x_0}(s)$.

F is continuous by the pasting lemma. (check the formulas agree on overlap $s = \frac{1}{2} - \frac{1}{2}t$, $s = \frac{1}{2} + \frac{1}{2}t$ - they all give $\gamma(1-t)$.)

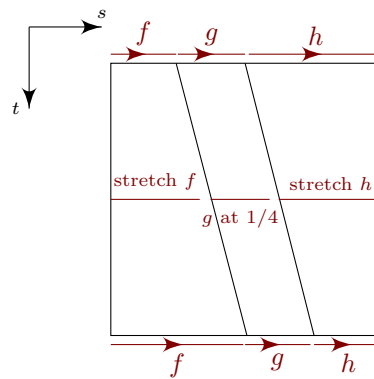
(iii) ASSOCIATIVITY. i.e. for all loops $f, g, h : I \rightarrow X$ based at x_0 , we have Lecture 27

$$([f] * [g]) * [h] = [f] * ([g] * [h])$$

in other words, we have

$$(f * g) * h \sim f * (g * h)$$

Make a homotopy $F : I \times I \rightarrow X$ mapping $(s, t) \mapsto F(s, t)$ between $(f * g) * h$ and $f * (g * h)$.



Define the homotopy

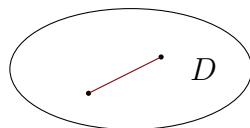
$$F(s, t) = \begin{cases} f\left(\frac{4s}{t+1}\right) & 0 \leq s \leq \frac{t+1}{4} \\ g(4s - (t+1)) & \frac{t+1}{4} \leq s \leq \frac{t+2}{4} \\ h\left(\frac{4}{2-t}\left[s - \frac{t+2}{4}\right]\right) & \frac{t+2}{4} \leq s \leq 1 \end{cases}$$

(iv) CLOSURE: Obvious so we'll ignore it. ■

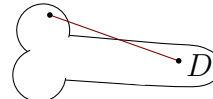
8.1 Computing $\pi_1(X, x_0)$ in simple cases

Definition (Convex). A subset $D \subseteq \mathbb{R}^n$ is called *convex* if and only if whenever $x_0, x_1 \in D$, the line segment from x_0 to x_1 is also in D .

E.g. A convex subset:



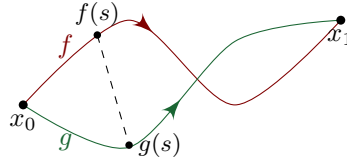
not convex¹:



¹but mildly humorous.

Claim: Let $D \subseteq \mathbb{R}^n$ be convex and suppose that $f, g : I \rightarrow D$ are paths in D from x_0 to x_1 . Then $f \sim g$. [In a convex domain, any two paths with the same endpoints are homotopic].

Proof. We need to construct a homotopy from f to g .



The straight line from $f(s)$ to $g(s)$ is always in D , so it makes sense to define $F : I \times I \rightarrow D$ mapping $(s, t) \mapsto (1 - t)f(s) + tg(s)$. F is continuous (as it is the sum of products of continuous functions). and

$$F(s, 0) = f(s) \text{ for all } s \in I$$

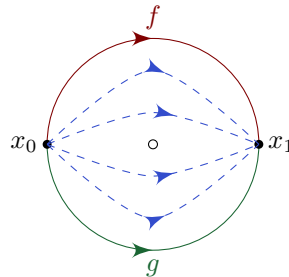
$$F(s, 1) = g(s) \text{ for all } s \in I$$

$$F(0, t) = x_0 \text{ for all } t \in I$$

$$F(1, t) = x_1 \text{ for all } t \in I$$

which together imply that F is a homotopy from f to g . ■

Example 8.3. $X = \mathbb{R}^2 \setminus \{0\}$.



f is *not* homotopic to g (to be proved later...)

Corollary 8.4. Let D be a convex subset of $\mathbb{R}^n$, and $x_0 \in D$. Then

$$\pi_1(D, x_0) = \{1\}$$

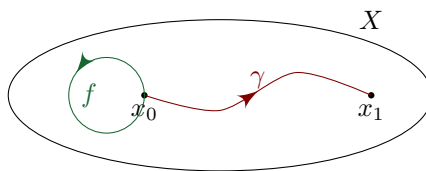
Proof. Any loop γ based at x_0 is homotopic to e_{x_0} . ■

Definition (Simply Connected). If $\pi_1(X, x_0) = \{1\}$ for all $x_0 \in X$, we say that X is *simply connected*.

Recall from the problem sheet, a domain is star shaped if we have a line from x to x_0 which lies in D , and star-shaped domains are simply-connected.

Dependence on Basepoint

How is $\pi_1(X, x_0)$ related to $\pi_1(X, x_1)$? See problem sheet.



Suppose that $\exists$ a path γ from x_0 to x_1 . Define

$$M_\gamma : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$$

mapping $[f] \mapsto [\gamma^{-1} * f * \gamma]$, a loop based at x_1 .

Then M_γ is well-defined, and M_γ is a group homomorphism:

$$M_\gamma([f] * [g]) = [\gamma^{-1} * f * g * \gamma]$$

and

$$\begin{aligned} M_\gamma([f]) * M_\gamma([g]) &= [\gamma^{-1} * f * \gamma] * [\gamma^{-1} * g * \gamma] \\ &= [\gamma^{-1} * f * \gamma * \gamma^{-1} * g * \gamma] \end{aligned}$$

So $M_\gamma([f] * [g]) = M_\gamma([f]) * M_\gamma([g])$. Also $(M_\gamma)^{-1} = M_{\gamma^{-1}}$. So M_γ is an isomorphism of groups. So, up to isomorphism, $\pi_1(X, x_0)$ depends only on the path component of X containing x_0 , and not on x_0 itself.

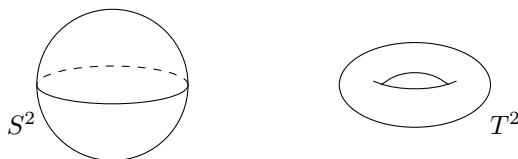
Question: What happens to the fundamental group when we apply f ?

I claim that $f : \pi_1(X, x_0) \rightarrow \pi_1(Y, f(x_0))$ maps $[\gamma] \mapsto [f \circ \gamma]$.

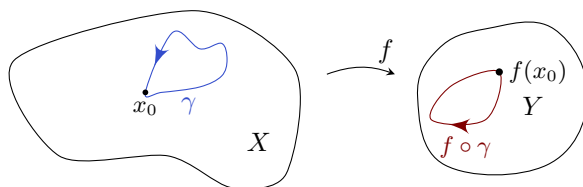
Motivating Question

Is S^2 homeomorphic to the 2-dimensional torus, T^2 ?

Lecture 28



Why does the fundamental group help here? Given topological spaces X, Y and $f : X \rightarrow Y$ continuous,



Whenever $\gamma : I \rightarrow X$ is a loop based at $x_0 \in X$, $f \circ \gamma : I \rightarrow Y$ is a loop based at $f(x_0)$. Similarly, whenever $F : I \times I \rightarrow X$ is a homotopy between two loops (or paths), $\gamma_0, \gamma_1 : I \rightarrow X$ in X , then $f \circ F$ is a homotopy from $f \circ \gamma_0$ to $f \circ \gamma_1$: $F : I \times I \rightarrow X$ is a homotopy from γ_0 to γ_1 , mapping $(s, t) \mapsto F(s, t)$ iff F is continuous and

$$\begin{aligned} F(s, 0) &= \gamma_0(s) \text{ for all } s \in I \\ F(s, 1) &= \gamma_1(s) \text{ for all } s \in I \\ F(0, t) &= x_0 \text{ for all } t \in I \\ F(1, t) &= x_1 \text{ for all } t \in I \end{aligned}$$

Thus $f : X \rightarrow Y$ induces a well-defined map, $f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, f(x_0))$ mapping $[\gamma] \mapsto [f \circ \gamma]$. Note that $f_*([\gamma_0] * [\gamma_1])$ is the homotopy class of loops with one representative given by the loop based at $f(x_0)$ in Y given by $I \rightarrow Y$, where

$$s \mapsto \begin{cases} f \circ \gamma_0(2s) & 0 \leq s \leq \frac{1}{2} \\ f \circ \gamma_1(2s - 1) & \frac{1}{2} \leq s \leq 1 \end{cases}$$

So

$$f_*([\gamma_0] * [\gamma_1]) = (f_*[\gamma_0]) * (f_*[\gamma_1])$$

So f_* is a group homomorphism.

Summarising: Given a topological space X and a point $x_0 \in X$, we get a group $\pi_1(X, x_0)$ whenever $f : X \rightarrow Y$ is continuous, we get a group homomorphism $f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, f(x_0))$. Furthermore, if $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then

$$\begin{array}{ccccc} \pi_1(X, x_0) & \xrightarrow{f_*} & \pi_1(Y, f(x_0)) & \xrightarrow{g_*} & \pi_1(Z, g(f(x_0))) \\ & & \searrow & \nearrow & \\ & & (g \circ f)_* & & \end{array}$$

Indeed

$$g_*(f_*[\gamma]) = g_*([f \circ \gamma]) = [g \circ f \circ \gamma] = (g \circ f)_*[\gamma]$$

So $(g \circ f)_* = g_* \circ f_*$.

Theorem 8.5

Let X, Y be topological spaces and $f : X \rightarrow Y$ be a homeomorphism. Then

$$f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, f(x_0))$$

is an isomorphism.

Proof. f is a homeomorphism, so $\exists g : Y \rightarrow X$ continuous such that $f \circ g = \text{id}_Y$ and $g \circ f = \text{id}_X$. I claim that $g_* : \pi_1(Y, f(x_0)) \rightarrow \pi_1(X, x_0)$ is the inverse to f_* . Because $\text{id} = (\text{id}_Y)_* = (f \circ g)_* = f_* \circ g_*$. Similarly, $\text{id} = (\text{id}_X)_* = (g \circ f)_* = g_* \circ f_*$. So $f_* \circ g_* = \text{id}$ and $g_* \circ f_* = \text{id}$. ■

8.2 Homotopy of Maps

Idea: Two maps $f_0, f_1 : X \rightarrow Y$ are called *homotopic* if I can find a “movie of maps” $f_t : X \rightarrow Y, 0 \leq t \leq 1$, interpolating continuously between f_0 and f_1

Definition (Homotopic Maps). Continuous maps $f_0, f_1 : X \rightarrow Y$ are *homotopic* if and only if $\exists F : X \times I \rightarrow Y$ mapping $(x, t) \mapsto F(x, t)$ such that $F(x, 0) = f_0(x)$ for all $x \in X$ and $F(x, 1) = f_1(x)$ for all $x \in X$.

Definition (Basepoint-preserving). Given $x_0 \in X$ we say that such a homotopy $F : X \times I \rightarrow Y$ mapping $(x, t) \mapsto F(x, t)$ is *basepoint-preserving* iff $F(x_0, t) = f(x_0)$ for all $t \in I$.

Lecture 29

Note that if there is a basepoint-preserving homotopy from f to g then $f(x_0) = g(x_0)$, where $f, g : X \rightarrow Y$.

Notation. A *space with basepoint* (X, x_0) is a topological space X together with a point $x_0 \in X$. A *continuous map* $f : (X, x_0) \rightarrow (Y, y_0)$ between spaces with basepoint is a continuous $f : X \rightarrow Y$ such that $f(x_0) = y_0$.

Proposition 8.6. Let $f, g : (X, x_0) \rightarrow (Y, y_0)$ be continuous maps that are homotopic via a basepoint-preserving homotopy. Then

$$f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$$

and

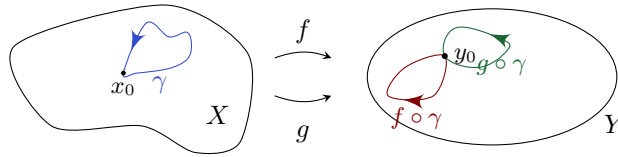
$$g_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$$

are equal.

Proof. Note that

$$\begin{aligned} f_* = g_* &\iff f_*[\gamma] = g_*[\gamma] \text{ for all loops } \gamma \text{ based at } x_0 \in X \\ &\iff f \circ \gamma \sim g \circ \gamma \text{ for all loops } \gamma \text{ based at } x_0 \in X \end{aligned}$$

We have the basepoint-preserving $[F(x_0, t) = y_0, \forall t \in I]$ homotopy $F : X \times I \rightarrow Y$ mapping $(x, t) \mapsto F(x, t)$ continuous, and have $\gamma : I \rightarrow X$ mapping $s \mapsto \gamma(s)$.



Consider $H : I \times I \rightarrow Y$ mapping $(s, t) \mapsto F(\gamma(s), t)$. [“follow γ along the movie of map F ”]. Then H is continuous (composition of continuous functions) and

$$\begin{aligned} H(s, 0) &= F(\gamma(s), 0) = f \circ \gamma(s) \\ H(s, 1) &= F(\gamma(s), 1) = g \circ \gamma(s) \\ H(0, t) &= F(\gamma(0), t) = F(x_0, t) = y_0 \quad \forall t \in I \\ H(1, t) &= y_0, \quad \forall t \in I \end{aligned}$$

So H is a homotopy from $f \circ \gamma$ to $g \circ \gamma$, so $f_* = g_*$ as claimed. ■

Aside: Recall that if $f : X \rightarrow Y$ is a homeomorphism then $f_* = \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$ is an isomorphism of groups

$$X \xrightarrow{f} Y \xrightarrow{g} X$$

where $g_* \circ f_* = (g \circ f)_* = (\text{id}_X)_* = \text{id}$ etc. So f_* is an isomorphism.

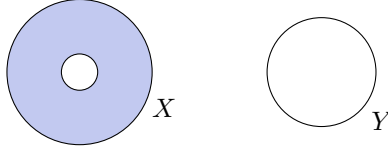
Definition (Homotopy Equivalence). A continuous map $f : (X, x_0) \rightarrow (Y, y_0)$ is called a basepoint-preserving *homotopy equivalence* if and only if $\exists g : (Y, y_0) \rightarrow (X, x_0)$ continuous such that $g \circ f \sim \text{id}_X$, $f \circ g \sim \text{id}_Y$.

Example 8.7. If $f : (X, x_0) \rightarrow (Y, y_0)$ is a homeomorphism then it is a homotopy equivalence.

Example 8.8. Take

$$X = \{(x, y) \in \mathbb{R}^2 : 1 \leq x^2 + y^2 \leq 4\}, \quad x_0 = (\sqrt{3}, 0)$$

$$Y = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 3\}, \quad y_0 = (\sqrt{3}, 0)$$



Then certainly

$$Y \xrightarrow{i} X \xrightarrow{\pi} Y$$

where in polar coordinates π maps $(r, \theta) \mapsto (\sqrt{3}, \theta)$.

Then $\pi \circ i : (Y, y_0) \rightarrow (Y, y_0)$ is the identity map. But $i \circ \pi : (X, x_0) \rightarrow (X, x_0)$ mapping $(r, \theta) \mapsto (\sqrt{3}, \theta)$.

Claim: $i \circ \pi \sim \text{id}_X$.

Proof. Take the homotopy $F : X \times I \rightarrow X$ sending

$$((r, \theta), t) \mapsto (rt + (1 - t)\sqrt{3}, \theta)$$

F is continuous, and

$$F((r, \theta), 0) = (\sqrt{3}, \theta) = i \circ \pi((r, \theta))$$

$$F((r, \theta), 1) = (r, \theta) = \text{id}_X((r, \theta))$$

■

Thus $\pi_1(X, x_0) \cong \pi_1(Y, y_0)$ because whenever $f : (X, x_0) \rightarrow (Y, y_0)$ is a homotopy equivalence, $f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$ is an isomorphism.

Example 8.9. $\mathbb{R}^2 \setminus \{0\} \sim \text{circle}$. (same argument as previous example).

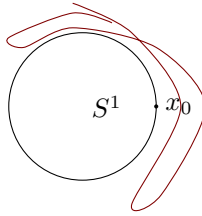
8.3 Fundamental Group of the Circle

Theorem 8.10

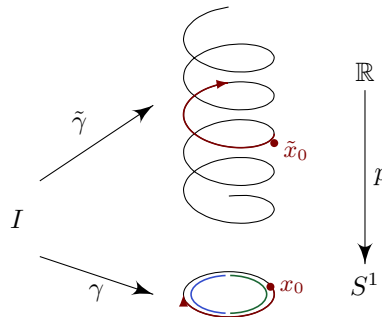
$$\pi_1(S^1, x_0) \cong \mathbb{Z}.$$

Key notion: winding number.

Suppose that $\gamma : I \rightarrow S^1$ is a loop based at $x_0 \in S^1$. We will define an integer, the *winding number* of γ , which is the net number of times that γ winds around S^1 . E.g. below the winding number is 0, “total change in angle around the loop is zero”:



Consider $p : \mathbb{R} \rightarrow S^1$ sending $t \mapsto (\cos 2\pi t, \sin 2\pi t)$, $x_0 = (1, 0)$.



Lemma 8.11 (Path lifting). *For any path $\gamma : I \rightarrow S^1$ starting at $x_0 \in S^1$ and any $\tilde{x}_0 \in \mathbb{R}$ such that $p(\tilde{x}_0) = x_0$ there is a unique path $\tilde{\gamma} : I \rightarrow \mathbb{R}$ such that $\tilde{\gamma}$ starts at $\tilde{x}_0$ and $p \circ \tilde{\gamma} = \gamma$.*

How to apply this: Given a loop $\gamma : I \rightarrow S^1$ based at $x_0 = (1, 0)$. Take $\tilde{x}_0 = 0 \in \mathbb{R}$ and let $\tilde{\gamma} : I \rightarrow \mathbb{R}$ be the lift of γ . Then $\tilde{\gamma}(1) \in \mathbb{Z}$. (because $p \circ \tilde{\gamma} = \gamma$, so $p(\tilde{\gamma}(1)) = \gamma(1) = (1, 0)$, and $p^{-1}(1, 0) = \mathbb{Z} \subset \mathbb{R}$.)

Lecture 30

Definition (Winding Number). Define $w(\gamma)$, the *winding number* of γ , to be the integer $\tilde{\gamma}(1)$.

Theorem 8.12

The map $\pi_1(S^1, x_0) \rightarrow \mathbb{Z}$ sending $[\gamma] \mapsto w(\gamma)$ is well defined and an isomorphism of groups.

Proof. From the problem sheet, we had the lemma:

Lemma 8.13 (Homotopy lifting). *For any homotopy $F : I \times I \rightarrow S^1$ and any $\tilde{x}_0$ such that $p(\tilde{x}_0) = F(0, 0)$, there exists a unique homotopy $\tilde{F} : I \times I \rightarrow \mathbb{R}$ such that $p \circ \tilde{F} = F$.*

To show that our map $\pi_1(S^1, x_0) \rightarrow \mathbb{Z}$ is well-defined, suppose that γ_1 and γ_2 are loops in S^1 based at x_0 and that $F : I \times I \rightarrow S^1$ is a homotopy between them. Applying the lemma above gives $\tilde{F} : I \times I \rightarrow \mathbb{R}$ such that $p \circ \tilde{F} = F$. Let $\tilde{\gamma}_0 : I \rightarrow \mathbb{R}$ be the unique lift of γ_0 starting at $\tilde{x}_0$, and let $\tilde{\gamma}_1 : I \rightarrow \mathbb{R}$ be the unique lift of γ_1 starting at $\tilde{x}_0$.

We need to show $w(\gamma_0) = w(\gamma_1)$ or in other words, $\tilde{\gamma}_0(1) = \tilde{\gamma}_1(1)$. The map $I \rightarrow \mathbb{R}$ sending $s \mapsto \tilde{F}(s, 0)$ is a path in $\mathbb{R}$ starting at $\tilde{x}_0$ satisfying $p \circ \tilde{\gamma}_0$. So by uniqueness, this path is $\tilde{\gamma}_0$. Similarly $\tilde{F}(s, 1) = \tilde{\gamma}_1(s)$ for all $s \in I$. So $\tilde{F}$ is a homotopy from $\tilde{\gamma}_0$ to $\tilde{\gamma}_1$. In particular, therefore, $\tilde{\gamma}_0(1) = \tilde{\gamma}_1(1)$ and so our map $\pi_1(S^1, x_0) \rightarrow \mathbb{Z}$ mapping $[\gamma] \mapsto w(\gamma)$ is well-defined.

It remains to show the map is

- surjective
- injective
- a group homomorphism

SURJECTIVITY: Given $n \in \mathbb{Z}$, we need to find $\gamma : I \rightarrow S^1$, a loop based at x_0 such that $w(\gamma) = n$. Take $\gamma(s) = (\cos(2\pi ns), \sin(2\pi ns))$. Then $\tilde{\gamma}(s) = ns$ satisfies $p \circ \tilde{\gamma} = \gamma$, and so $\tilde{\gamma}$ is the unique lift of γ starting at $0 \in \mathbb{R}$, thus $w(\gamma) = \tilde{\gamma}(1) = n$.

INJECTIVITY: We need to show that if γ_0, γ_1 satisfy $w(\gamma_0) = w(\gamma_1)$, then $\gamma_0 \sim \gamma_1$ as loops in S^1 . But write $\tilde{\gamma}_0, \tilde{\gamma}_1 : I \rightarrow \mathbb{R}$ for the lifts of γ_0, γ_1 that start at $0 \in \mathbb{R}$. Then $\tilde{\gamma}_0(1) = \tilde{\gamma}_1(1)$. And $\mathbb{R}$ is convex, so any two paths with the same endpoints are homotopic. Let $F : I \times I \rightarrow \mathbb{R}$ be a homotopy from $\tilde{\gamma}_0$ to $\tilde{\gamma}_1$. Then $p \circ F$ is a homotopy from γ_0 to γ_1 .

GROUP HOMOMORPHISM: Left as an *exercise*.

Thus $\pi_1(S^1, x_0) \cong \mathbb{Z}$. ■

Motivating Question: Is S^2 homeomorphic to $S^1 \times S^1$?

Lecture 31

No: If S^2 were homeomorphic to $S^1 \times S^1$, then $\pi_1(S^2, x_0) \cong \pi_1(S^1 \times S^1, y_0)$. But

$$\pi_1(X \times Y, (a, b)) \cong \pi_1(X, a) \times \pi_1(Y, b)$$

Aside: Why is this true? Product topology on $X \times Y$ satisfies

$$\begin{array}{ccc} & & X \\ & \nearrow p_X & \\ Z & \xrightarrow{F} & X \times Y \\ & \searrow p_Y & \\ & & Y \end{array}$$

The universal property says $Z \xrightarrow{F} X \times Y$ is continuous if and only if $p_X \circ F$ and $p_Y \circ F$ are both continuous. More concretely, to give $F : Z \rightarrow X \times Y$ sending $z \mapsto (F_1(z), F_2(z))$ is the same as giving $F_1 : Z \rightarrow X$ and $F_2 : Z \rightarrow Y$ and F is continuous iff both F_1 and F_2 are continuous.

A path $\gamma : I \rightarrow X \times Y$ sending $s \mapsto (\gamma_1(s), \gamma_2(s))$ is the same thing as to give $\gamma_1 : I \rightarrow X, \gamma_2 : I \rightarrow Y$ paths in X and Y . Take the map

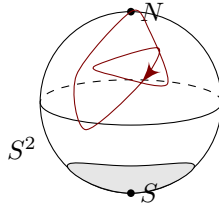
$$\begin{aligned} G : \pi_1(X \times Y, (a, b)) &\longrightarrow \pi_1(X, a) \times \pi_1(Y, b) \\ [\gamma] &\mapsto ([\gamma_1], [\gamma_2]) \end{aligned}$$

and a homotopy $H : I \times I \rightarrow X \times Y$ sending $(s, t) \mapsto (H_1(s, t), H_2(s, t))$ is the same thing as to give two homotopies $H_1 : I \times I \rightarrow X, H_2 : I \times I \rightarrow Y$. After some thinking, we get if γ is homotopic to γ' , then γ_1 is homotopic to γ'_1 and γ_2 is homotopic to γ'_2 . So the map G is well-defined. Actually it's an isomorphism (mock exam question). Thus $\pi_1(S^1 \times S^1, y_0) \cong \mathbb{Z} \times \mathbb{Z}$.

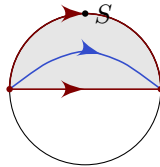
Claim: $\pi_1(S^2, x_0) \cong \{1\}$. S^2 is simply-connected.

Proof. We need to show that every loop in S^2 based at x_0 is homotopic to a constant loop at x_0 . Take $x_0 = N$, the north pole.

- (i) Step 1: γ is homotopic to a loop which does not pass through the South pole S . i.e.

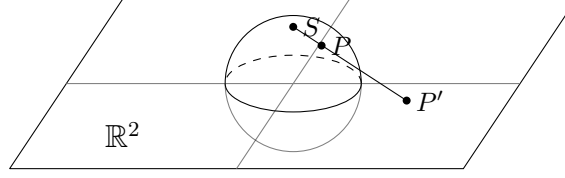


Sketch of proof. Let D be a small disc containing S but not containing N . Then every time γ passes through S , we have



Homotope that part of γ to go round the edge of D . Do this without moving $\gamma(I) \cap \partial D$. So every time we hit the south pole we can homotope off S .

- (ii) Step 2: $S^2 \setminus \{S\}$ is simply-connected: Using stereographic projection, we see that $S^2 \setminus \{S\} \cong \mathbb{R}^2$.



Consider $f : S^2 \setminus \{S\} \rightarrow \mathbb{R}^2$ where $x \mapsto$ the unique point of $\mathbb{R}^2$ where the straight line from S to x cuts $\mathbb{R}^2$. [Gap of boring computation] This is a continuous bijection. So $S^2 \setminus \{S\} \cong \mathbb{R}^2$ and $\pi_1(S^2 \setminus \{S\}, x'_0) \cong \pi_1(\mathbb{R}^2, y'_0) \cong \{1\}$.

So every loop in $S^2 \setminus \{S\}$ is homotopic to a constant loop. In particular, γ is homotopic to a constant loop as claimed. So $\pi_1(S^2, x_0) = \{1\}$. ■

Corollary 8.14. $S^2 \not\cong S^1 \times S^1$.

Felina. Finally we prove the fundamental theorem of algebra:

Theorem 8.15: Fundamental Theorem of Algebra

Let $p \in \mathbb{C}[z]$ be a degree n polynomial. Then p has n roots in $\mathbb{C}$, counted with multiplicity.

Proof. It suffices to prove that if $\deg p > 0$ then p has a root in $\mathbb{C}$ [division algorithm: $p(z) = (z - a)q(z) + r, r \in \mathbb{C}, r = p(a)$.]

Suppose that p has no roots in $\mathbb{C}$. Consider the path $\gamma_r : I \rightarrow S^1$ mapping

$$s \mapsto \frac{p(re^{2\pi is})}{p(r)} \cdot \frac{|p(r)|}{|p(re^{2\pi is})|}$$

γ_r is a loop in S^1 based at $1 \in S^1$. As r varies, these loops are homotopic. So $\gamma_r \sim \gamma_0 = \text{constant loop}$. Now imagine taking $r \gg 0$. Writing $p(z) = z^n + a_{n-1}z^{n-1} + \dots + a_0$, so for r sufficiently large

$$|z^n| > |a_{n-1}z^{n-1} + \dots + a_0|$$

Fix this r . Then for all $t \in [0, 1]$, the polynomial

$$p_t(z) = z^n + t(a_{n-1}z^{n-1} + \dots + a_0)$$

has no roots on $|z| = r$. So the loops in S^1 , $\gamma_{r,t} : I \rightarrow S$ sending

$$s \mapsto \frac{p_t(re^{2\pi is})}{p_t(r)} \cdot \frac{|p_t(r)|}{|p_t(re^{2\pi is})|}$$

are all homotopic. So $\gamma_0 \sim \gamma_r = \gamma_{r,1} \sim \gamma_{r,0} = e^{2\pi ins}$. So $n = 0$. Thus if $p \in \mathbb{C}[z]$ is a polynomial with no roots in $\mathbb{C} \implies \deg p = 0$. ■

- End of Metric Spaces & Topology -